BESOV-ISH SPACES THROUGH ATOMIC DECOMPOSITION: THE LEAN 4 FORMALIZATION

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Abstract. This document describes, without proofs, exactly those results of the paper *Besov-ish spaces through atomic decomposition* (Analysis & PDE 15 (2022), no. 1) that have been formally verified in the Lean 4 repository `BesovSpacesGoodGrid`, together with a few results obtained in the formalization that go beyond the paper. It was produced from a prerelease version of the published paper (`preprint-source/besovish-2020-07-18.tex` in the repository), but all references to numbered statements of the paper follow the *published* version. Unlike the paper, the document is organized to mirror the repository: Part I covers the weak-grid library (`WeakGrid/`) and Part II the good-grid library (`GoodGrid/`), with sections ordered by file dependency; each section opens with a box naming the Lean file it documents and the project files that file imports. Statements follow the notation of the paper; each result is followed, in the running text, by a *Formalization* paragraph naming the corresponding Lean declarations, describing what each of them proves, and recording every deviation from the published statements, together with a snapshot of the Lean code. The document also explains the purpose and architecture of the formalization and ends with a guide to the repository files. The repository compiles with no `sorry`, and the main theorems depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

Contents

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1. Introduction

1.1. **Purpose of the formalization.** The repository `BesovSpacesGoodGrid` formalizes, in Lean 4 over mathlib, a substantial part of the atomic decomposition theory of [1]. The goals are twofold. First, to obtain machine-checked proofs of the main results of the paper — the Banach structure, compactness and completeness of Besov-ish spaces, the transmutation-of-atoms machinery, the equivalence of the atomic, Haar, and mean-oscillation norms, and the theory of pointwise multipliers up to the non-Archimedean property and its positive-cone refinements. Second, and just as important, to produce a *reusable* Lean library: the basic theory is stated and proved for abstract *weak grids* and abstract atom families, and the good-grid / Souza-atom spaces of the paper are then obtained as a specialization. The weak-grid layer (grids, atoms, Besov-ish spaces, scales, completeness, transmutation) is independent of the good-grid geometry and is intended to be usable in later developments.

This document is organized to mirror the repository rather than the paper. Part I documents the weak-grid library (`BesovSpacesGoodGrid/WeakGrid/`) and Part II the good-grid library (`BesovSpacesGoodGrid/GoodGrid/`); inside each part the sections follow the import order of the Lean files, so that every section only uses material from earlier sections. Each section opens with a box naming the file it documents and the project files it imports. The mathematical pipeline, in this order, is:

(1) define quantitative good grids and general weak grids;
(2) package weak-grid cells and local spaces, and define atom families with support, convexity, phase invariance, and atom-size estimates;
(3) build levelwise atomic blocks and infinite L^p representations, define the Besov-ish space and the coefficient-cost gauge `Norm_Costpq`;
(4) prove the L^t embedding, scale inclusions, representation-limit theorems, compactness statements, and completeness for the cost norm;
(5) prove the transmutation claims moving atomic representations from one weak grid to another, and specialize them to compare Souza's atoms, Besov's atoms, and sandwiched atom families on good grids;
(6) introduce the positive cone, the Haar and standard representation gauges, and the mean-oscillation norm, with finite-constant comparison theorems;
(7) develop the pointwise-multiplier layer: the `selfs` classes, boundedness of `selfs` multipliers, the $\mathcal{B}^s_{1,1}$ multiplier characterization, and the non-Archimedean multiplier estimates, including their positive-cone versions.

A guide to the repository files is given in Section 19.

1.2. **How this document was produced.** This is a mathematical description of the current state (June 9, 2026) of the Lean 4 formalization of [1]. It was produced from a *prerelease version of the published paper* (the LaTeX source `preprint-source/besovish-2020-07-18.tex`, included in the repository) by the following rules.

- Results that are not present in the repository were deleted. These are: the elementary estimates of paper Section 4 (Hölder-like and convolution tricks, inlined in the Lean proofs without standalone statements); the Hölder and bounded variation atoms (paper Sections 11B–11C) and the corresponding characterizations (paper Propositions 16.2 and 16.3); Dirac's approximations (paper Section 17); Corollary 18.6, strongly regular domains (paper 18.7–18.8, except for the grid-cell case, which is formalized), Pointwise Multipliers I and II (paper 18.9–18.10); and all of paper Sections 19–21 (quasialgebra, description of $\mathcal{B}^s_{1,1}$ by indicators, left compositions).

- Statements were updated to match the precision of the formalization; every change with mathematical content is recorded in the *Formalization* paragraph that follows the statement.
- All references of the form "paper Proposition $x.y$" use the numbering of the *published* version, Analysis & PDE **15** (2022), no. 1 (which the reader should consult alongside this document); the prerelease source in the repository has the same section structure but its statement numbers may differ.
- The material was reordered to follow the repository (see above), so the section numbers of this document do *not* match the section numbers of the paper.
- No proofs are included. Each statement is followed by a *Formalization* paragraph giving the corresponding Lean declarations and files, with a short description of what each declaration proves, and then by a snapshot of the Lean code that formalizes it (the formal statement as it appears in the repository, with the proof omitted; an ellipsis ... marks the omitted proof or, in a few long definitions, omitted structural fields).

A global change must be singled out: *throughout this document* $p \in [1, \infty)$, $q \in [1, \infty]$ *and* $u \in [1, \infty]$. The paper develops Part I for the quasi-Banach range $p, q \in (0, \infty]$; the formalization restricts to the Banach range, so $\rho = \min\{1, p, q\} = 1$ everywhere and all the ρ-norms of the paper are genuine norms. In Lean the standing hypotheses are `1 <= p`, `p != ∞`, `Fact (1 <= q)`, `1 <= u`, `0 < s`.

2. Notation

We removed from the paper's table the rows for Hölder atoms, bounded variation atoms, the constants ρ and $\hat{t}$ (no longer needed in the Banach range) since the corresponding material is not formalized. Constants denoted C, $C(\cdot)$ below depend only on the displayed arguments; the Lean statements carry explicit values for most of them.

I. THE WEAK-GRID LIBRARY

`BesovSpacesGoodGrid/WeakGrid/`

In Part I we assume $s > 0$, $p \in [1, \infty)$, $u \in [1, \infty]$ and $q \in [1, \infty]$.[a]

[a] Paper: $s > 0$, $p \in (0, \infty)$, $q \in (0, \infty]$. See the Introduction for the restriction to the Banach range.

Table 1. Most important notation/symbols/constants

Symbol	Description
I	phase space
m	finite measure in the phase space I
a_P, b_P	an atom supported on P
$\mathcal{A}$	a class of atoms
$\mathcal{A}(Q)$	set of atoms of class $\mathcal{A}$ supported on Q
$\mathcal{A}^{sz}_{s,p}$	class of (s,p)-Souza's atoms
$\mathcal{A}^{bs}_{s,\beta,p,q}$	class of (s,β,p,q)-Besov's atoms
$\mathcal{P}$	grid of I
$\mathcal{P}^k$	family of subsets of I at the k-th level of $\mathcal{P}$
$\lambda_1 \leq \lambda_2$	describes geometry of the good grid $\mathcal{P}$
$\mathcal{B}^s_{p,q}(\mathcal{A})$	(s,p,q)-Banach space defined by the class of atoms $\mathcal{A}$
$\mathcal{B}^s_{p,q}$ or $\mathcal{B}^s_{p,q}(\mathcal{A}^{sz}_{s,p})$	(s,p,q)-Banach space defined by Souza's atoms
P, Q, W	elements of the grid $\mathcal{P}$
L^p or $L^p(m)$	Lebesgue spaces of $(I,\mathbb{A},m)$
$\lvert \cdot \rvert_p$	norm in L^p, $p \in (0,\infty]$
p'	$1/p + 1/p' = 1$, with $p \in [1,\infty]$

3. Measure spaces and weak grids

File: `WeakGrid/Definition.lean`
Imports (within the project): `none (external: UnbalancedHaarWavelet, LaminarFamiliesMaximalBinaryTrees)`

The starting point of the whole library is deliberately poor in structure: no metric, no topology, no nesting between scales — only a finite measure space together with a sequence of finite families of measurable sets ("cells"), one family per scale k, satisfying a single quantitative hypothesis: a uniform bound C_1 on how many cells of the *same* level can meet a given cell. This bounded-overlap condition G_1 is exactly what is needed to make levelwise ℓ^p bookkeeping work: when a level-k sum of locally supported functions is integrated, each point is covered by at most C_1 summands, so L^t norms of levelwise sums can be compared with ℓ^p norms of their coefficients at the cost of powers of C_1 (this is the computation behind inequality (6-5) of the paper). Everything in Part I — the Besov-ish spaces, their completeness, compactness, and the transmutation machinery — is built on this hypothesis alone.

Let I be a measure space with a σ-algebra $\mathbb{A}$ and m be a measure on $(I,\mathbb{A})$, $m(I) < \infty$. Given a measurable set $J \subset I$ denote $|J| = m(J)$. We denote the Lebesgue spaces of $(I,\mathbb{A},m)$ by L^p. A **weak grid** is a sequence of finite families of measurable sets with positive measure $\mathcal{P} = (\mathcal{P}^k)_{k\in\mathbb{N}}$, so that at least one of these families is non empty and

G_1. Given $Q \in \mathcal{P}^k$, let
$$\Omega_Q^k = \{P \in \mathcal{P}^k\colon\ P \cap Q \neq \emptyset\}.$$
Then
$$C_1 = \sup_k \sup_{Q\in\mathcal{P}^k} \#\Omega_Q^k < \infty.$$

Define $|\mathcal{P}^k| = \sup\{|Q|\colon Q \in \mathcal{P}^k\}$. We often abuse notation using $\mathcal{P}$ for both $(\mathcal{P}^k)_{k\in\mathbb{N}}$ and $\cup_k \mathcal{P}^k$.

Formalization. A grid is the structure `WeakGrid` in `WeakGrid/Definition.lean` (fields `partitions`, `measurable`, `positive_measure`, `exists_nonempty`, `Cmult1`, `overlap_card_le`), bundled with its ambient measurable space as `WeakGridSpace`. The paper calls this simply a "grid"; in the repository it is called a *weak grid*, to distinguish it from the good grids of Section 11. The paper's convention that $P \neq Q$ for cells at different levels is replaced by indexing cells by (level, cell) pairs (the structure `WeakGridCell`). Implementation notes: each level is a `Finset (Set α)`, so all levelwise sums are finite sums and no convergence issues arise inside a level; the exponents p, q, u live in $\mathbb{R}_{\geq 0} \cup \{\infty\}$ (the type `ENNReal`), which lets the endpoint $q = \infty$ or $u = \infty$ be handled by the same statements instead of by parallel "usual modification" versions; and the overlap neighborhood Ω_Q^k is the definition `overlapFinset`, on which G_1 is imposed as a cardinality bound.

```
structure WeakGrid where
        /-- The measure on ‘α‘. -/
        μ : MeasureTheory.Measure α
        /-- The measure is finite on the whole space. -/
        isFinite : MeasureTheory.IsFiniteMeasure μ
        /-- The finite family ‘P^k‘ at each level ‘k‘. -/
        partitions : ℕ → Finset (Set α)
        /-- Every cell is measurable. -/
        measurable : ∀ k Q, Q ∈ partitions k → MeasurableSet Q
        /-- Every cell has positive measure. -/
        positive_measure : ∀ k Q, Q ∈ partitions k → 0 < μ Q
        /-- At least one level is nonempty. -/
        exists_nonempty : ∃ k, (partitions k).Nonempty
        /-- Uniform same-level overlap multiplicity bound. -/
        Cmult1 : ℕ
        /-- ‘# Ω_Q^k ≤ Cmult1‘ for every ‘Q ∈ P^k‘. -/
        overlap_card_le :
          ∀ k Q, Q ∈ partitions k → (overlapFinset (partitions k)
    Q).card ≤ Cmult1

structure WeakGridSpace where
        grid : WeakGrid (α := α)
```

Remark 3.1. There are plenty of examples of spaces with grids; the simplest one is $[0,1)$ with the Lebesgue measure and the dyadic grid $\mathcal{D}^k = \{[i/2^k, (i+1)/2^k),\ 0 \le i < 2^k\}$, and similarly the dyadic and d-adic grids of $[0,1)^n$. Any measure space with a finite non-atomic measure can be endowed with a grid made of a nested sequence of partitions whose elements at the same level have precisely the same measure.

4. Atoms

File: `WeakGrid/Atoms.lean`
Imports (within the project): `WeakGrid/Definition.lean`

An atom family assigns to every cell Q a set $\mathcal{A}(Q)$ of model functions supported on Q, normalized so that an atom of a cell of measure $|Q|$ has size at most $|Q|^{s-1/(u'p)}$ in L^{pu}. The normalization is chosen so that the smoothness parameter s plays, on an abstract measure space, the role that derivative order plays on $\mathbb{R}^n$: smaller cells must carry atoms with larger norm before they contribute unit cost, exactly as in the classical atomic decompositions of Besov spaces. Two structural axioms deserve comment. Convexity and phase invariance of $\mathcal{A}(Q)$ (A_3) are what make the infimum norm of Section 5 a genuine norm: given representations of f and of g, the triangle inequality is proved by merging them cellwise, and the merged "atom" is a convex combination of phase-rotated atoms, hence again an atom (this is the proof of paper Proposition 6.3). The optional compactness axiom A_5 is what later turns boundedness in cost into compactness in L^p. The flexibility of the abstraction — a possibly different local model space $\mathcal{B}(Q)$ for every cell — is used in earnest in Part II, where the same interface is instantiated by one-dimensional spaces of constants (Souza's atoms) and by infinite-dimensional induced Besov spaces (Besov's atoms).

Let $\mathcal{P}$ be a grid. Let $p \in [1,\infty)$, $u \in [1,\infty]$, and $s > 0$. A family of **atoms** associated with $\mathcal{P}$ of type (s,p,u) is an indexed family $\mathcal{A}$ of pairs $(\mathcal{B}(Q), \mathcal{A}(Q))_{Q \in \cup_k \mathcal{P}^k}$ where

- A_1. $\mathcal{B}(Q)$ is a complex linear space of functions contained in L^{pu}.
- A_2. If $\phi \in \mathcal{B}(Q)$ then $\phi(x) = 0$ for every $x \notin Q$.
- A_3. $\mathcal{A}(Q)$ is a nonempty convex subset of $\mathcal{B}(Q)$ such that $\phi \in \mathcal{A}(Q)$ implies $\sigma\phi \in \mathcal{A}(Q)$ for every $\sigma \in \mathbb{C}$ satisfying $|\sigma| = 1$.
- A_4. We have
$$|\phi|_{pu} \le |Q|^{s-\frac{1}{u'p}}$$
for every $\phi \in \mathcal{A}(Q)$, where $1/u + 1/u' = 1$.

We will say that $\phi \in \mathcal{A}(Q)$ is an $\mathcal{A}$-atom of type (s,p,u) supported on Q. Sometimes we also assume

A_5. For every $Q \in \mathcal{P}$ we have that $\mathcal{A}(Q)$ is a compact subset in the strong topology of L^p,

or

A_6. every $\mathcal{A}(Q)$ is a sequentially compact subset in the weak topology of L^p.

Formalization. Atom families are the structure `AtomFamily` in `WeakGrid/Atoms.lean`, whose fields formalize A_1–A_4. The local space $\mathcal{B}(Q)$ is the structure `LocalVectorSpace` (field `localSpace`): in the paper A_1 asks for a complex *Banach* space contained in L^{pu}, while Lean only requires a complex vector space together with an injective linear map into the functions $\alpha \to \mathbb{C}$, all whose members belong to L^{pu} (field `local_memLp`); no completeness of the local space is assumed, and it is never used. A_2 is the field `local_support`; A_3 corresponds to `atoms_nonempty`, `atoms_convex` and `atoms_phase_invariant` (the paper states the phase invariance as an "if and only if"; the two formulations are equivalent); A_4 is the field `atom_bound`. The compactness assumption A_5 appears as the predicate `AssumptionA5`. The paper's alternative assumption A_7 ($\mathcal{B}(Q)$ finite dimensional and $\mathcal{A}(Q)$ closed) is used in the paper only to derive A_5; in the formalization the finite-dimensional case is instantiated directly for Souza's atoms. Implementation notes: the Hölder conjugate u' is a structure field (`uConj`) constrained by the mathlib-style predicate `ENNReal.HolderConjugate`, rather than a computed value, which avoids case splits at $u \in \{1, \infty\}$; the local space construction `LocalVectorSpace` is universe-polymorphic, so the same `AtomFamily` interface accepts both small concrete models (the constants $\mathbb{C}$, for Souza's atoms) and large function-space models (induced L^p spaces, for Besov's atoms); and the atom-size bound is stated with `MeasureTheory.eLpNorm`, the $\mathbb{R}_{\geq 0} \cup \{\infty\}$-valued L^p seminorm of mathlib, so no integrability side conditions are needed to state A_4.

```
structure AtomFamily
  (G : WeakGridSpace (α := α)) (s : ℝ) (p u : ℝ≥0∞) where
  /-- The Hölder conjugate exponent ‘u’‘. -/
  uConj : ℝ≥0∞
  /-- The smoothness parameter is positive. -/
  s_pos : 0 < s
  /-- ‘p ∈ [1,∞)‘. -/
  one_le_p : 1 ≤ p
  /-- ‘p < ∞‘, expressed in ‘ℝ≥0∞‘ as ‘p ≠ ∞‘. -/
  p_ne_top : p ≠ ∞
  /-- ‘u ∈ [1,∞]‘. -/
  one_le_u : 1 ≤ u
  /-- ‘uConj‘ is the Hölder conjugate of ‘u‘. -/
  holder_conjugate : ENNReal.HolderConjugate u uConj
  /-- The local vector space ‘B(Q)‘. -/
  localSpace : WeakGridCell G → LocalVectorSpace.{u, v} α
  /-- The chosen atoms ‘A(Q)‘, as elements of ‘B(Q)‘. -/
  atoms : ∀ Q, Set ((localSpace Q).carrier)
  /-- Every cell has at least one atom. -/
  atoms_nonempty : ∀ Q, (atoms Q).Nonempty
```

```
/-- `B(Q)` is contained in `L^{pu}`. -/
local_memLp : ∀ Q φ, MeasureTheory.MemLp ((localSpace Q).toFun φ) (p * u)
  G.measure
/-- Local functions are supported on `Q`. -/
local_support : ∀ Q φ, ∀ x, x ∉ Q.cell → (localSpace Q).toFun φ x = 0
/-- `A(Q)` is convex in the local vector space. -/
atoms_convex : ∀ Q, Convex ℝ (atoms Q)
/-- `A(Q)` is invariant under multiplication by complex scalars of modulus one. -/
atoms_phase_invariant :
  ∀ Q φ (σ : ℂ), φ ∈ atoms Q → ‖σ‖ = (1 : ℝ) → σ • φ ∈ atoms Q
/-- The atom size estimate in `L^{pu}`. -/
atom_bound :
  ∀ Q φ, φ ∈ atoms Q →
    MeasureTheory.eLpNorm ((localSpace Q).toFun φ) (p * u) G.measure ≤
      atomMeasureScale G s p uConj Q
```

5. Besov-ish spaces

File: `WeakGrid/BesovishSpaces.lean`
Imports (within the project): `WeakGrid/Atoms.lean`

This is the core file of the library. The Besov-ish space is defined by *declaring* what an atomic decomposition is and taking the functions that admit one: a representation is a choice, for every level k and cell Q, of a coefficient s_Q and an atom a_Q, such that the levelwise sums converge to g in L^p; its *cost* is the ℓ^q norm over levels of the ℓ^p norms over the cells of each level — precisely the sequence-space norm of the classical Frazier–Jawerth atomic decomposition, transplanted to an abstract grid. Since representations are far from unique, the norm is the infimum of the cost over all of them. Two analytic inputs drive the section. The bounded overlap G_1 gives the levelwise estimate (6-5) of the paper, and the summability hypothesis G_2 (a weighted geometric-type series over the levels) lets the levelwise estimates be summed; together they yield the embedding $\mathcal{B}^s_{p,q}(\mathcal{A}) \subset L^t$ of Proposition 5.1, which in particular ($t = p$) shows that a finite-cost series really does define an element of L^p and that the infimum norm separates points. Convexity and phase invariance of the atom sets then give the triangle inequality, so the cost gauge is a norm.

Let $p \in [1, \infty)$, $u \in [1, \infty]$, $q \in [1, \infty]$, $s > 0$, $\mathcal{P} = (\mathcal{P}^k)_{k \geq 0}$ be a grid and let $\mathcal{A}$ be a family of atoms of type (s, p, u). We will also assume

G_2. We have

$$\Big(\sum_k |\mathcal{P}^k|^{sq^*}\Big)^{1/q^*} < \infty \quad \text{(usual modification if } q^* = \infty), \qquad \lim_k |\mathcal{P}^k| = 0,$$

where q^* is the exponent produced by the Hölder-like estimate of the paper (for $q > 1$, $q^* = q'$).

Formalization. G_2 is the predicate `AssumptionG2` in `WeakGrid/BesovishSpaces.lean`, combining the finiteness of the coefficient-weight series needed for the $t = p$ embedding with $\lim_k |\mathcal{P}^k| = 0$. The paper writes this condition as $C(p, q, (|\mathcal{P}^k|^s)_k) < \infty$ using the constant of its Proposition 4.1; since paper Section 4 has no standalone Lean counterpart, the formalization states the required summability of the weighted level measures directly.

The **Besov-ish space** $\mathcal{B}^s_{p,q}(I, \mathcal{P}, \mathcal{A})$ is the set of all complex valued functions $g \in L^p$ that can be represented by an absolutely convergent series in L^p

$$g = \sum_{k=0}^{\infty} \sum_{Q \in \mathcal{P}^k} s_Q a_Q \tag{1}$$

where a_Q is in $\mathcal{A}(Q)$, $s_Q \in \mathbb{C}$ and with finite **cost**

$$\Big(\sum_{k=0}^{\infty} \big(\sum_{Q \in \mathcal{P}^k} |s_Q|^p\big)^{q/p}\Big)^{1/q} < \infty \tag{2}$$

(usual modification when $q = \infty$). The series in (1) is called a **$\mathcal{B}^s_{p,q}(I, \mathcal{P}, \mathcal{A})$-representation** of the function g. Define

$$|g|_{\mathcal{B}^s_{p,q}(I,\mathcal{P},\mathcal{A})} = \inf \Big(\sum_{k=0}^{\infty} \big(\sum_{Q \in \mathcal{P}^k} |s_Q|^p\big)^{q/p}\Big)^{1/q},$$

where the infimum runs over all possible representations of g as in (1).

Formalization. All in `WeakGrid/BesovishSpaces.lean`. For each level k, the structure `LevelBlock` chooses one coefficient and one atom per cell of $\mathcal{P}^k$; its value in L^p (the finite sum $\sum_{Q \in \mathcal{P}^k} s_Q a_Q$) is `LevelBlock.toLp`. A representation (1) is the structure `LpGridRepresentation`: a sequence of level blocks together with a `HasSum` witness in L^p. The cost (2) is `pqCost` (with the $\mathbb{R}_{\geq 0} \cup \{\infty\}$-valued variant `pqCostENNReal`), and its finiteness is the predicate `FinitePQCost`, with the endpoint $q = \infty$ represented by a supremum bound. Membership with some finite-cost representation is `MemBesovishCoeffCost`; `BesovishSpace` is the corresponding `Submodule` of L^p; and the norm $|\cdot|_{\mathcal{B}^s_{p,q}(I,\mathcal{P},\mathcal{A})}$ is the cost gauge `Norm_Costpq`, an infimum over representation costs. Implementation notes: convergence of the series (1) is expressed by mathlib's `HasSum` over the level index, inside the Banach space `Lp` (over $\mathbb{C}$, exponent p), which encapsulates the paper's "absolutely convergent in L^p" once the levelwise sums are grouped into blocks; the hypotheses $1 \leq p$, $1 \leq q$ are carried as `Fact` instances so that mathlib's L^p machinery picks them up by instance resolution; and the normed structure on the space is provided as a *local* instance (`costNormedAddCommGroup`) rather than a global one, because the same underlying L^p element may belong to several Besov-ish spaces with different gauges.

When the choice of I and/or $\mathcal{P}$ is obvious we write $\mathcal{B}^s_{p,q}(\mathcal{P}, \mathcal{A})$ or $\mathcal{B}^s_{p,q}(\mathcal{A})$; whenever we write just $\mathcal{B}^s_{p,q}$ we choose the Souza's atoms $\mathcal{A}^{sz}_{s,p}$ of Section 12.

```
structure LpGridRepresentation
    (A : AtomFamily G s p u) (g : Lp ℂ p G.measure) where
  block : (k : ℕ) → LevelBlock A k
  hasSum : HasSum (fun k => (block k).toLp A) g

def BesovishSpace (A : AtomFamily G s p u) (q : ℝ≥0∞)
    [Fact (1 ≤ q)]
    : Submodule ℂ (Lp ℂ p G.measure) where
  -- Carrier: all ‘L^p‘ elements admitting a Besov-ish atomic
    representation with finite p q cost.
  carrier := { g | MemBesovishCoeffCost A q g }
  ...

noncomputable def Norm_Costpq
    (A : AtomFamily G s p u) (q : ℝ≥0∞) [Fact (1 ≤ q)]
    (g : BesovishSpace A q) : ℝ :=
  pqPseudoNorm A q g
```

Proposition 5.1 (Embedding into L^t). *Assume G_1–G_2 and A_1–A_4. Let $t \in [p, \infty)$ be such that*

$$s - \frac{1}{p} + \frac{1}{t} \geq 0, \qquad p \leq t \leq pu, \tag{3}$$

and suppose

$$C_2(t) = C_1^{1+1/t}\, C\big(t, q, (|\mathcal{P}^k|^{t(s-\frac{1}{p}+\frac{1}{t})})_k\big) < \infty. \tag{4}$$

Then for every g with a $\mathcal{B}^s_{p,q}(\mathcal{A})$-representation (1) of cost K the series converges in L^t and

$$|g|_t \leq C_2(t)\, K. \tag{5}$$

In particular

$$|g|_t \leq C_2(t)\, |g|_{\mathcal{B}^s_{p,q}(\mathcal{A})},$$

and $\mathcal{B}^s_{p,q}(\mathcal{A}) \subset L^t$ continuously.

Formalization. The theorem `lp_embedding_adapted_statement` in `WeakGrid/BesovishSpaces.lean` proves the per-representation inequality (5): for any finite-cost representation R of g, the L^t norm of g is bounded by the explicit constant $C_1^{1+1/t}$·`cCoefficient` times the cost of R. The norm inequality follows by taking the infimum over representations. Compared with paper Proposition 6.1 the hypotheses are identical except for $q \geq 1$ and $t \geq p$ being imposed globally, and the levelwise estimate (6.2) of the paper is internal to the proof rather than part of the statement.

```
theorem lp_embedding_adapted_statement
    {A : AtomFamily G s p u} {t : ℝ≥0∞}
    [Fact (1 ≤ t)]
```

```
    (hp_ne_top : p ≠ ∞) (ht_ne_top : t ≠ ∞)
    (hq_one : 1 ≤ q)
    (hp_le_t : p ≤ t) (ht_le_pu : t ≤ p * u)
    (hs_nonneg : 0 ≤ s - 1 / p.toReal + 1 / t.toReal)
    {g : Lp ℂ p G.measure} (R : LpGridRepresentation A g)
    (hRfin : LpGridRepresentation.FinitePQCost (q := q) R)
    (hCco_fin : cCoefficientFinite t q (fun k =>
      (levelMeasureWeight G s p t k) ^ t.toReal)) :
        (MeasureTheory.eLpNorm (g : α → ℂ) t G.measure).toReal ≤
        ((G.grid.Cmult1 : ℝ) ^ (1 + 1 / t.toReal)) *
        cCoefficient t q (fun k => (levelMeasureWeight G s p t k) ^
    t.toReal) *
          LpGridRepresentation.pqCost (q := q) R := ...
```

Proposition 5.2 (Linear structure)**.** *Assume G_1–G_2 and A_1–A_4. Then $\mathcal{B}^s_{p,q}(\mathcal{A})$ is a complex linear subspace of L^p and $|\cdot|_{\mathcal{B}^s_{p,q}(\mathcal{A})}$ is a norm on it. Moreover the inclusion*

$$\imath\colon (\mathcal{B}^s_{p,q}(\mathcal{A}), |\cdot|_{\mathcal{B}^s_{p,q}(\mathcal{A})}) \to (L^p, |\cdot|_p)$$

is continuous.

Formalization. `BesovishSpace` is by construction a `Submodule` of L^p, which gives the linear structure. The normed structure is the definition `BesovishSpace.costNormedAddCommGroup` (requiring $p \neq \infty$ and G_2), which installs `Norm_Costpq` as a norm — nonnegativity, triangle inequality and homogeneity are packaged, together with the L^t embedding estimate, in the summary theorem `BesovishSpace.normedSpace_and_lp_embedding`. Continuity of $\imath$ is the case $t = p$ of Proposition 5.1. Compared with paper Proposition 6.3, $\rho = \min\{1, p, q\} = 1$ since $p, q \geq 1$, so the ρ-norm of the paper is a norm.

```
noncomputable def costNormedAddCommGroup
    (hp_top : p ≠ ∞)
    (hCco_fin : LpGridRepresentation.cCoefficientFinite p q (fun k =>
      (LpGridRepresentation.levelMeasureWeight G s p p k) ^ p.toReal)) :
    NormedAddCommGroup (BesovishSpace A q) where
  norm := Norm_Costpq A q
  dist x y := Norm_Costpq A q (-x + y)
  ...
```

6. Limits of representations, compactness, and completeness

File: `WeakGrid/Completeness.lean`
Imports (within the project): `WeakGrid/Atoms.lean`, `WeakGrid/BesovishSpaces.lean`

The compactness and completeness theory rests on a single mechanism, the *limit of representations*: if functions g_n have representations with uniformly bounded cost, one extracts — by a Cantor diagonal argument over the countably many cells — a subsequence along which every coefficient converges and, using the compactness of the atom sets (A_5 or A_6), every atom converges as well; the limiting coefficients and atoms then form a representation of the limit function, and lower semicontinuity of the cost under cellwise convergence gives the same cost bound C. This one proposition yields, in cascade: sequential compactness of cost balls in the L^p topology, completeness of the Besov-ish space (a Cauchy sequence converges in L^p by the embedding, and the limit stays in the space with the right norm), and later all the good-grid compactness statements of Part II. A technical point that the formalization makes explicit: to even *state* the limit representation one must know that a block sequence with finite abstract cost is summable in L^p; this is `formalBlockSeq_summable`, proved from G_2, and it is the reason G_2 appears in all the statements of this section.

Proposition 6.1 (Limits of representations)**.** *Assume G_1–G_2 and A_1–A_4. Suppose that g_n are functions in $\mathcal{B}^s_{p,q}(\mathcal{A})$ with $\mathcal{B}^s_{p,q}(\mathcal{A})$-representations*

$$g_n = \sum_{k=0}^{\infty} \sum_{Q\in\mathcal{P}^k} s^n_Q a^n_Q$$

satisfying

i. *There is C such that for every n*

$$\Big(\sum_{k=0}^{\infty}\Big(\sum_{Q\in\mathcal{P}^k} |s^n_Q|^p\Big)^{q/p}\Big)^{1/q} \le C.$$

ii. *For every $Q \in \mathcal{P}$ the limit $s_Q = \lim_n s^n_Q$ exists.*
iii. *For every $Q \in \mathcal{P}$ there is $a_Q \in \mathcal{A}(Q)$ such that either $a^n_Q \to a_Q$ in the strong topology of L^p, or $a^n_Q \to a_Q$ weakly in L^p.*

Then g_n converges (strongly, respectively weakly) in L^p to a function $g \in \mathcal{B}^s_{p,q}(\mathcal{A})$ which has the $\mathcal{B}^s_{p,q}(\mathcal{A})$-representation

$$g = \sum_{k=0}^{\infty} \sum_{Q\in\mathcal{P}^k} s_Q\, a_Q$$

of cost at most C.

Formalization. In `WeakGrid/Completeness.lean`, matching paper Proposition 6.4. The theorem `representation_limit` proves the weak version: under uniform cost bounds (i), cellwise convergence of the coefficients (ii) and weak convergence of the atoms (iii), the limiting blocks define a Besov-ish function of cost at most C and the g_n converge weakly. The theorem `representation_limit_strong` is the strong L^p

version: strong convergence of the atoms gives strong convergence of the represented functions.

```
theorem representation_limit
    (A : AtomFamily G s p u)(hG2 : AssumptionG2 G s p u q)
     {gseq : ℕ → Lp ℂ p G.measure}
    {gLim : Lp ℂ p G.measure} {C : ℝ}
    (H : RepresentationLimitHypotheses A q gseq gLim C)
    (hp_ne_top : p ≠ ∞) (hs_pos : 0 < s) (hu_one : 1 ≤ u)
    [Fact (1 ≤ u)] (hC : 0 ≤ C) :
    MemBesovishCoeffCost A q gLim ∧
      LpGridRepresentation.FinitePQCost (q := q) H.Rlim ∧
      LpGridRepresentation.pqCost (q := q) H.Rlim ≤ C ∧
      Tendsto (fun n => toWeakSpace ℂ (Lp ℂ p G.measure) (gseq n)) atTop
        (𝓝 (toWeakSpace ℂ (Lp ℂ p G.measure) gLim)) := ...
```

Corollary 6.2 (Compactness and completeness)**.** *Assume G_1–G_2, A_1–A_4, and either A_5 or A_6. Then*

i. *If $g_n \in \mathcal{B}^s_{p,q}(\mathcal{A})$ satisfy $|g_n|_{\mathcal{B}^s_{p,q}(\mathcal{A})} \leq C$ for every n, then there is a subsequence converging (strongly under A_5, weakly under A_6) in L^p to some $g \in \mathcal{B}^s_{p,q}(\mathcal{A})$ with $|g|_{\mathcal{B}^s_{p,q}(\mathcal{A})} \leq C$.*

ii. *Under A_5, every closed ball of $(\mathcal{B}^s_{p,q}(\mathcal{A}), |\cdot|_{\mathcal{B}^s_{p,q}(\mathcal{A})})$ is sequentially compact in the strong topology of L^p.*

iii. *Under A_5, $(\mathcal{B}^s_{p,q}(\mathcal{A}), |\cdot|_{\mathcal{B}^s_{p,q}(\mathcal{A})})$ is a complex Banach space.*

Formalization. All in `WeakGrid/Completeness.lean`. Item i. is `representation_limit_strong_existence` and `representation_limit_weak_existence` (paper Corollary 6.5(i)): they start from a bare sequence of limit blocks, prove it is summable, and package the resulting limit as a genuine representation of cost at most C. Item ii. is `exists_strongly_convergent_subseq_of_uniform_pqCost` (extraction of a strongly convergent subsequence from any uniformly cost-bounded sequence; the extended version also returns the convergence of the coefficients and of the atoms) and `closed_Norm_Costpq_ball_strongly_seqCompact` (sequential compactness of closed cost balls in the strong L^p topology); paper Corollary 6.5(iii) states the compactness of the inclusion $\imath$ under A_5, and the formalization states the sequential compactness of closed cost balls, which is the equivalent form used downstream. Item iii. is `besovishSpace_costNorm_completeSpace`, completeness of the Besov-ish space for the cost norm (paper Corollary 6.5(ii); for $p, q \geq 1$ the ρ-Banach space of the paper is a Banach space, and paper Corollary 6.7 is the finite-dimensional criterion through which the Souza case of Part II is obtained). The paper's Corollary 6.6 (existence of a representation attaining the infimum cost) is not formalized and is omitted.

```
theorem closed_Norm_Costpq_ball_strongly_seqCompact
    (hp_ne_top : p ≠ ∞) (hs_pos : 0 < s) (hu_one : 1 ≤ u)
    (A : AtomFamily G s p u) (hG2 : AssumptionG2 G s p u q)
    (hA5 : AssumptionA5 A)
    {C : ℝ}
    [Fact (1 ≤ u)]
    (hC : 0 ≤ C)
    (gseq : ℕ → BesovishSpace A q)
    (hball : ∀ n, BesovishSpace.Norm_Costpq A q (gseq n) ≤ C) :
    ∃ (φ : ℕ → ℕ) (_hφ : StrictMono φ) (gLim : BesovishSpace A q),
      BesovishSpace.Norm_Costpq A q gLim ≤ C ∧
        Tendsto
          (fun n => ((gseq (φ n) : BesovishSpace A q) : Lp ℂ p G.measure))
          atTop
          (𝓝 ((gLim : BesovishSpace A q) : Lp ℂ p G.measure)) := ...

theorem besovishSpace_costNorm_completeSpace
    (hp_ne_top : p ≠ ∞) (hs_pos : 0 < s) (hu_one : 1 ≤ u)
    (A : AtomFamily G s p u) (hG2 : AssumptionG2 G s p u q)
    (hA5 : AssumptionA5 A)
    [Fact (1 ≤ u)] :
    @CompleteSpace (BesovishSpace A q)
      (BesovishSpace.costNormedAddCommGroup
        (A := A) (q := q) hp_ne_top
    hG2.1).toMetricSpace.toPseudoMetricSpace.toUniformSpace := ...
```

7. Scales of spaces

File: `WeakGrid/Scales.lean`
Imports (within the project): `WeakGrid/BesovishSpaces.lean`

Changing the smoothness parameter from s to $\tilde{s}$ amounts to rescaling every atom of a cell Q by the deterministic factor $|Q|^{\tilde{s}-s}$ — there is no analysis in the rescaling itself, only in comparing the resulting spaces. When $s \le \tilde{s}$ and the level measures decay fast enough (a weighted summability over levels, the same shape as G_2), a representation in the finer scale converts levelwise into a representation in the coarser scale with a summable loss, giving the continuous inclusion $\mathcal{B}^{\tilde{s}}_{p,\tilde{q}} \subset \mathcal{B}^{s}_{p,q}$ — the abstract analogue of the classical monotonicity of Besov scales in the smoothness parameter.

A family of atoms $\mathcal{A}$ of type (s, p, u) induces a one-parameter scale $\tilde{s} \mapsto \mathcal{A}_{\tilde{s},p}$, where $\mathcal{A}_{\tilde{s},p}$ is the family of atoms of type $(\tilde{s}, p, u)$ defined by

$$\mathcal{A}_{\tilde{s},p}(Q) = \{|Q|^{\tilde{s}-s} a_Q \colon\ a_Q \in \mathcal{A}(Q)\}.$$

Proposition 7.1 (Scale inclusion)**.** *Assume G_1–G_2 and that $\mathcal{A}$ satisfies A_1–A_4. Let $0 < s \le \tilde{s}$ and $q, \tilde{q} \in [1, \infty]$. Suppose*

$$\Big(\sum_k |\mathcal{P}^k|^{q^*(\tilde{s}-s)}\Big)^{1/q^*} < \infty.$$

Then $\mathcal{B}^{\tilde{s}}_{p,\tilde{q}}(\mathcal{A}_{\tilde{s},p}) \subset \mathcal{B}^{s}_{p,q}(\mathcal{A}_{s,p})$, the inclusion is a continuous linear map, and

$$|g|_{\mathcal{B}^s_{p,q}(\mathcal{A}_{s,p})} \le C\big(q, (|\mathcal{P}^k|^{\tilde{s}-s})_k\big)\, |g|_{\mathcal{B}^{\tilde{s}}_{p,\tilde{q}}(\mathcal{A}_{\tilde{s},p})}.$$

Formalization. In `WeakGrid/Scales.lean`. The cellwise rescaling by the deterministic factor $\mu(Q)^{\tilde{s}-s}$ is implemented by `smoothnessScaleFactor`, `smoothnessScaleAtoms` and `smoothnessScaleAtomFamily`. The theorem `smoothnessScaleBesovishSpace_subset` proves the set inclusion under the deterministic summability hypothesis `scaleCoefficientFinite` and $s \le \tilde{s}$; the definition `smoothnessScaleBesovishSpaceInclusion` packages it as a linear map, and the theorem `smoothnessScaleBesovishSpaceInclusion_Norm_Costpq_le` gives the norm bound with the explicit constant `scaleCoefficient`. This is item A of paper Proposition 7.1; items B (compactness of bounded sets of the finer space in the coarser norm) and C (compactness of the inclusion under A_7) are not formalized and are omitted. For $u = \infty$ the two-parameter scaling map `smoothnessIntegrabilityScaleAtomFamily` (scale factor $\mu(Q)^{\tilde{s}-s+1/p-1/\tilde{p}}$) is also defined, but the paper's two-parameter scale inclusion is not formalized.

```
theorem smoothnessScaleBesovishSpace_subset
  [Fact (1 ≤ p)]
  [Fact (1 ≤ u)] [Fact (1 ≤ q)] [Fact (1 ≤ qTilde)]
    (A : AtomFamily G s p u) (sTilde : ℝ) (hsTilde_pos : 0 < sTilde)
    (hs_le : s ≤ sTilde)
    (hqfin : scaleCoefficientFinite q (smoothnessScaleLevelWeight G s
    sTilde p))
    {g : Lp ℂ p G.measure}
    (hg : g ∈ SmoothnessScaleBesovishSpace
      (A.smoothnessScaleAtomFamily sTilde hsTilde_pos) qTilde) :
    g ∈ BesovishSpace A q := ...
```

8. Induced spaces

File: `WeakGrid/InducedGrid.lean`
Imports (within the project): `WeakGrid/BesovishSpaces.lean`

Induced spaces formalize the idea of *zooming into a cell*: fixing a cell Q at level k_0, the cells of deeper levels contained in Q form a grid of the measure space $(Q, m|_Q)$, with levels re-indexed to start at 0, and the atoms supported in Q form an atom family on this induced grid. Reading a representation on the induced grid as a representation on the ambient grid costs nothing (the coefficients are the same, placed at levels $k_0 + i$), which is why the inclusion below is a weak contraction. This construction is the basic localization tool of the library: Besov's atoms (Section 14) are *defined* through induced norms, and the multiplier theory (Section 18) constantly moves between a cell and the ambient space through it. The paper introduces induced spaces for good grids (Section 10 of the published version); the formalization develops them for arbitrary weak grids, since no good-grid geometry is needed.

Consider a Besov-ish space $\mathcal{B}^s_{p,q}(I, \mathcal{P}, \mathcal{A})$, where $\mathcal{P}$ is a grid. Given $Q \in \mathcal{P}^{k_0}$, consider the sequence of finite families of subsets $\mathcal{P}_Q = (\mathcal{P}^i_Q)_{i\geq 0}$ of Q given by

$$\mathcal{P}^i_Q = \{P \in \mathcal{P}^{k_0+i},\ P \subset Q\},$$

and let $\mathcal{A}_Q$ be the restriction of $\mathcal{A}$ to indices in $\mathcal{P}_Q$. Then we can consider the **induced** Besov-ish space $\mathcal{B}^s_{p,q}(Q, \mathcal{P}_Q, \mathcal{A}_Q)$, and the inclusion

$$i \colon \mathcal{B}^s_{p,q}(Q, \mathcal{P}_Q, \mathcal{A}_Q) \to \mathcal{B}^s_{p,q}(I, \mathcal{P}, \mathcal{A})$$

is well defined and a weak contraction:

$$|f|_{\mathcal{B}^s_{p,q}(I,\mathcal{P},\mathcal{A})} \leq |f|_{\mathcal{B}^s_{p,q}(Q,\mathcal{P}_Q,\mathcal{A}_Q)}.$$

Formalization. In `WeakGrid/InducedGrid.lean`. The induced grid and its ambient space are the definitions `inducedWeakGrid` and `inducedWeakGridSpace`; the restricted atom family is `inducedAtomFamily`; and the transfer of a representation from the induced grid to the ambient one, with control of cost (which gives the weak contraction above), is `inducedRepresentationToAmbient`. For Souza's atoms the formalization also transfers positivity along this map (`inducedRepresentationToAmbient_atom_toFunction_pos`); this is the form needed by the multiplier theory of Section 18. The reverse restriction map is treated, for cells of the grid, in Lemma 18.1 below.

```
def inducedRepresentationToAmbient
    (G : WeakGridSpace (α := α)) {k₀ : ℕ} (Q : LevelCell G k₀)
    {s : ℝ} {p u : ℝ≥0∞} [Fact (1 ≤ p)] (A : AtomFamily G s p u)
    {f : Lp ℂ p (inducedWeakGridSpace G Q).measure}
    (R : LpGridRepresentation (inducedAtomFamily G Q A) f) :
    LpGridRepresentation A (inducedLpToAmbient G Q f) := ...
```

9. Transmutation of atoms

File: `WeakGrid/Transmutation.lean`
Imports (within the project): `WeakGrid/Atoms.lean, WeakGrid/BesovishSpaces.lean, WeakGrid/Completeness.lean`

The key result in Part I (paper Proposition 8.1) states that if we replace each atom of an atomic decomposition by a combination of atoms of a possibly different class, in such way that we do not change the location of the supports and the mass of the new representation concentrates pretty much on the original scale, then we obtain a new atomic decomposition whose cost is controlled by the cost of the original one.

The mechanism is worth describing, because the formalization follows it closely. Each source atom a_Q, $Q \in \mathcal{G}^i$, is expanded in target atoms $b_{P,Q}$ with coefficients $s_{P,Q}$ whose levelwise ℓ^p mass decays geometrically away from the level k_i attached to i (hypothesis III); the level selector $i \mapsto k_i$ is only assumed *almost linear* (hypothesis II), which is the right invariance for comparing two grids whose scales are exponentially related. Summing over Q with coefficients c_Q and collecting, for each target cell P, all contributions, one obtains target coefficients $m_P = \sum_Q |c_Q s_{P,Q}|$ and normalized target atoms d_P (a convex combination, by A_3 again). Estimating the cost of $(m_P)_P$ is then a discrete convolution inequality across levels: the geometric decay λ^{k-k_i} acts as a convolution kernel in the level variable, and the paper's convolution trick (Proposition 4.2) bounds the $\ell^{q/p}$ norm of the convolution by the kernel mass times the source cost. This is where the explicit constant $C_1 C_6^{1/p} \lambda^{-B/p} C_3(p,q,b) C_7^{1/q}$ of (8-17) comes from. The truncated sums (over source levels $i \leq N$) are exact identities of finite character (Claim I); each truncation is a genuine representation with the right cost (Claim II); and the passage $N \to \infty$ uses the limit-of-representations theorem of Section 6 (Claim III) — this is why the file imports `Completeness`.

Proposition 9.1 (Transmutation of atoms)**.** *Assume*

I. *$\mathcal{A}_2$ is a class of (s,p,u_2)-atoms for a grid $\mathcal{W}$, satisfying A_1–A_4 and G_1–G_2. Let $\mathcal{G}$ be also a grid satisfying G_1–G_2.*

II. *$k_i \in \mathbb{N}$, $i \geq 0$, is a sequence such that there are $\alpha > 0$ and $A, B \in \mathbb{R}$ with*

$$\alpha i + A \leq k_i \leq \alpha i + B \quad \text{for every } i.$$

III. *There is $\lambda \in (0,1)$ such that the following holds. For every $Q \in \mathcal{G}^i$ there are atoms $b_{P,Q} \in \mathcal{A}_2(P)$ and coefficients $s_{P,Q} \in \mathbb{C}$, indexed by the cells $P \in \mathcal{W}$ with $P \subset Q$, such that*

$$h_Q = \sum_k \sum_{P \in \mathcal{W}^k,\, P \subset Q} s_{P,Q} b_{P,Q}$$

is a $\mathcal{B}^s_{p,q}(\mathcal{A}_2)$-representation of a function h_Q, with $s_{P,Q} = 0$ whenever $P \in \mathcal{W}^k$ with $k < k_i$, and moreover

$$\sum_{P \in \mathcal{W}^k,\, P \subset Q} |s_{P,Q}|^p \leq C_3\, \lambda^{k-k_i} \quad \text{for every } k \geq k_i. \tag{6}$$

Then

A. *For all coefficients* $(c_Q)_{Q\in\mathcal{G}}$ *with*

$$\Big(\sum_i \Big(\sum_{Q\in\mathcal{G}^i} |c_Q|^p\Big)^{q/p}\Big)^{1/q} < \infty,$$

the sequence $N \mapsto \sum_{i\leq N}\sum_{Q\in\mathcal{G}^i} c_Q h_Q$ *converges in* L^p *to a function in* $\mathcal{B}^s_{p,q}(\mathcal{A}_2)$ *that has a* $\mathcal{B}^s_{p,q}(\mathcal{A}_2)$*-representation* $\sum_k\sum_P m_P d_P$ *with* $m_P \geq 0$ *and*

$$\Big(\sum_k \Big(\sum_{P\in\mathcal{W}^k} |m_P|^p\Big)^{q/p}\Big)^{1/q} \leq C_4 \Big(\sum_i \Big(\sum_{Q\in\mathcal{G}^i} |c_Q|^p\Big)^{q/p}\Big)^{1/q},$$

where $C_4 = C_4(C_1, C_3, \lambda, \alpha, A, B, p, q)$ *is explicit.*

B. *Suppose, in addition to the assumptions of A., that the* $s_{P,Q}$ *are nonnegative real numbers and* $b_{P,Q} > 0$ *on* P *for all* P, Q*. If* $m_P \neq 0$ *and* $d_P \neq 0$ *somewhere on* P*, then* $P \subset Q$ *for some cell* $Q \in \mathcal{G}$ *with* $c_Q \neq 0$ *and* $s_{P,Q} > 0$*, and the corresponding* h_Q *is a positive multiple of a positive function a.e. on* P*. If moreover* $c_Q \geq 0$ *for every* Q*, then* $m_P \neq 0$ *implies* $d_P > 0$ *a.e. on* P*.*

C. *Let* $\mathcal{A}_1$ *be a class of* (s, p, u_1)*-atoms for the grid* $\mathcal{G}$ *satisfying* A_1*–*A_4*. Suppose that for every atom* $a_Q \in \mathcal{A}_1(Q)$ *we can find* $s_{P,Q}$ *and* $b_{P,Q}$ *as in III. such that* $h_Q = a_Q$*. Then*

$$\mathcal{B}^s_{p,q}(\mathcal{A}_1) \subset \mathcal{B}^s_{p,q}(\mathcal{A}_2)$$

and this inclusion is continuous, with $|\phi|_{\mathcal{B}^s_{p,q}(\mathcal{A}_2)} \leq C_4\,|\phi|_{\mathcal{B}^s_{p,q}(\mathcal{A}_1)}$ *for every* $\phi \in \mathcal{B}^s_{p,q}(\mathcal{A}_1)$*.*

Formalization. In `WeakGrid/Transmutation.lean`. The transmuted objects (target coefficients $m_{P,N}$, normalized target atoms $d_{P,N}$, target level blocks, and their stable limits) are the definitions `TransmutationCoeff`, `TransmutationAtom`, `TransmutationBlock` and `TransmutationCoeffLimit`, `TransmutationAtomLimit`, `TransmutationBlockLimit`; the hypothesis III is the predicate `RepresentationWsubGandALS`, with the almost-linear level control II packaged as `AlmostLinearSequence`. The internal pipeline is formalized by the theorems `ClaimI` (exact bookkeeping identity: the transmuted representation at truncation N sums to the source partial expansion), `ClaimII` (each finite transmutation block sequence is a genuine representation of the source partial expansion, with cost controlled by the source cost) and `ClaimIII` (passage to the stable limiting data, producing the L^p limit and its representation), together with their endpoint versions `ClaimI_top`, `ClaimII_top`, `ClaimIII_top` for $q = \infty$. Item A is the theorem `Transmutation_of_Atoms_Claim_A`, with the explicit constant of the paper (product of C_1, $C_6^{1/p}$, $\lambda^{-B/p}$, the coefficient of the convolution trick — paper Proposition 4.2, which in Lean is the constant `cCoefficientInt` of the kernel `transmutationKernelZ` — and the integer-part factor $\max\{\ell \in \mathbb{N} : \ell < \alpha\} + 1$). Item B is `Transmutation_of_Atoms_Claim_B` and `Transmutation_of_Atoms_Claim_B_sharp` (support and positivity of the transmuted data); compared with paper Proposition 8.1.B, the pointwise positivity conclusions are stated almost everywhere on P (with positivity witnessed by positive real multiples), which is the natural formulation in L^p. Item C is `Transmutation_of_Atoms_Claim_C_explicit` and the continuous-embedding form `Transmutation_of_Atoms_continuous_embedding_explicit`. For the identity level selector, the theorem `LpGridRepresentation.cCoefficientInt_transmutationKernelZ_zero_one` computes the integer transmutation coefficient of the Claim C kernel as the geometric sum $(1-\rho)^{-1}$, $\lambda = \rho^p$; this is the constant simplification used in the Souza/Besov atom comparison of Section 14.

```
theorem Transmutation_of_Atoms_Claim_A
    (G W : WeakGridSpace (α := α))
    (AW : AtomFamily W s p u)
    (k : ℕ → ℕ)
    (A_als B_als r_als : ℝ)
    (hr_als : 0 < r_als)
    (hk_bound : ∀ i : ℕ,
      (k i : NNReal) ≤ r_als * (i : NNReal) + B_als ∧
      r_als * (i : NNReal) + A_als ≤ (k i : NNReal))
    (lam : ℝ) (hlam_pos : 0 < lam) (hlam_lt : lam < 1)
    (C : ℝ) (hC : 0 ≤ C)
    (h : (i : ℕ) → LevelCell G i → Lp ℂ p W.measure)
    (R : (i : ℕ) → (Q : LevelCell G i) → LpGridRepresentation AW (h i Q))
    (hR : RepresentationWsubGandALS (p := p) (q := q) G W AW k
      ⟨A_als, B_als, r_als, hr_als, hk_bound⟩ lam hlam_pos hlam_lt C hC h R)
    (c : (i : ℕ) → LevelCell G i → ℂ)
    (hc : CoeffFinitePQCost (p := p) (q := q) G c)
    (hq_ne_top : q ≠ ∞)
    (hG2_W : AssumptionG2 W s p u q)
    (hp_ne_top : p ≠ ∞)
    (hs_pos : 0 < s) :
    ∃ gLim : Lp ℂ p W.measure,
      ∃ hsum :
        HasSum (fun j => (TransmutationBlockLimit G W AW h R c A_als r_als j).toLp
     AW) gLim,
      MemBesovishCoeffCost AW q gLim ∧
      LpGridRepresentation.FinitePQCost (q := q)
        ({ block := TransmutationBlockLimit G W AW h R c A_als r_als
           hasSum := hsum } : LpGridRepresentation AW gLim) ∧
      Tendsto (fun N => PartialSumLevels G W h c N) atTop (𝓝 gLim) ∧
      LpGridRepresentation.pqCost (q := q)
        ({ block := TransmutationBlockLimit G W AW h R c A_als r_als
           hasSum := hsum } : LpGridRepresentation AW gLim) ≤
        (G.grid.Cmult1 : ℝ) *
        C ^ (1 / p.toReal) *
        lam ^ (-(B_als : ℝ) / p.toReal) *
        LpGridRepresentation.cCoefficientInt p ∞
          (transmutationKernelZ lam A_als r_als) *
        (Nat.ceil (r_als : ℝ) : ℝ) ^ (1 / q.toReal) *
        CoeffPQCost (p := p) (q := q) G c := ...

theorem Transmutation_of_Atoms_Claim_B (G W : WeakGridSpace (α := α))
    (AW : AtomFamily W s p u)
    (k : ℕ → ℕ)
```

```
    (A_als B_als r_als : ℝ)
    (hr_als : 0 < r_als)
    (hk_bound : ∀ i : ℕ,
      (k i : NNReal) ≤ r_als * (i : NNReal) + B_als ∧
      r_als * (i : NNReal) + A_als ≤ (k i : NNReal))
    (lam : ℝ) (hlam_pos : 0 < lam) (hlam_lt : lam < 1)
    (C : ℝ) (hC : 0 ≤ C)
    (h : (i : ℕ) → LevelCell G i → Lp ℂ p W.measure)
    (R : (i : ℕ) → (Q : LevelCell G i) → LpGridRepresentation AW (h i Q))
    (hR : RepresentationWsubGandALS_pos (p := p) (q := q) G W AW k
      ⟨A_als, B_als, r_als, hr_als, hk_bound⟩ lam hlam_pos hlam_lt C hC h R)
    (c : (i : ℕ) → LevelCell G i → ℂ)
    (hc : CoeffFinitePQCost (p := p) (q := q) G c)
    (hG2_W : AssumptionG2 W s p u q)
    (hp_ne_top : p ≠ ∞)
    (hs_pos : 0 < s) :
    (∃ gLim : Lp ℂ p W.measure,
      HasSum (fun j => (TransmutationBlockLimit G W AW h R c A_als r_als j).toLp
     AW) gLim ∧
      MemBesovishCoeffCost AW q gLim ∧
      Tendsto (fun N => PartialSumLevels G W h c N) atTop (𝒩 gLim)) ∧
    (∀ j : ℕ, ∀ P : LevelCell W j,
      TransmutationCoeffLimit G W AW h R c A_als r_als P ≠ 0 →
      (∃ x, x ∈ P.1 →
        AW.toFunction (levelCellToWeakGridCell W j P)
          (TransmutationAtomLocalLimit G W AW h R c A_als r_als P) x ≠ 0) →
        ∃ i ∈ Finset.range (transmutationStabilizationIndex A_als r_als j),
          ∃ Q ∈ (G.grid.partitions i).attach,
          P.1 ⊆ Q.1 ∧ c i Q ≠ 0 ∧
            ∃ r : NNReal, 0 < r ∧ ((R i Q).block j).coeff P = (r : ℂ) ∧
              ∀ᵐ x ∂ W.measure.restrict P.1,
                ∃ r : NNReal, 0 < r ∧ h i Q x = (r : ℂ)) := ...

theorem Transmutation_of_Atoms_continuous_embedding_explicit
    (G : WeakGridSpace (α := α))
    (u1 u2 : ℝ≥0∞)
    [Fact (1 ≤ u2)]
    (AG1 : AtomFamily G s p u1)
    (AG2 : AtomFamily G s p u2)
    (lam : ℝ) (hlam_pos : 0 < lam) (hlam_lt : lam < 1)
    (C : ℝ) (hC : 0 ≤ C)
    (hAG1_to_AG2 : ∀ i : ℕ, ∀ Q : LevelCell G i,
      ∀ g : Lp ℂ p G.measure,
        (∃ φ : (AG1.localSpace (levelCellToWeakGridCell G i Q)).carrier,
          AG1.IsAtom (levelCellToWeakGridCell G i Q) φ ∧
            g = atomLp AG1 (levelCellToWeakGridCell G i Q) φ) →
          ∃ Rg : LpGridRepresentation AG2 g,
            CoeffFinitePQCost (p := p) (q := q) G
              (fun j S => (Rg.block j).coeff S) ∧
            (∀ j : ℕ, ∀ S : LevelCell G j,
              (¬ S.1 ⊆ Q.1 → (Rg.block j).coeff S = 0) ∧
              (j < i → (Rg.block j).coeff S = 0)) ∧
            ∀ j : ℕ, i ≤ j → Rg.levelCoeffPower j ≤ C * lam ^ (j - i))
    (hG2_G : AssumptionG2 G s p u2 q)
    (hp_ne_top : p ≠ ∞)
```

```
(hs_pos : 0 < s) :
0 ≤ transmutationClaimCEmbeddingConstant G p q lam C ∧
  ∀ g : BesovishSpace AG1 q,
    ∃ hg2 : MemBesovishCoeffCost AG2 q (g : Lp ℂ p G.measure),
      BesovishSpace.Norm_Costpq AG2 q
          (⟨(g : Lp ℂ p G.measure), hg2⟩ : BesovishSpace AG2 q) ≤
        transmutationClaimCEmbeddingConstant G p q lam C *
          BesovishSpace.Norm_Costpq AG1 q g := ...
```

10. Further weak-grid infrastructure

Two more files complete the weak-grid layer; they contain no statement of the paper but are imported by Part II.

File: `WeakGrid/Multipliers.lean`
Imports (within the project): `WeakGrid/Completeness.lean, WeakGrid/InducedGrid.lean`

This file defines, at the weak-grid level of generality, the predicates on which the multiplier theory of Section 18 is built: a measurable g *is a pointwise multiplier* of a Besov-ish space (`IsPointwiseMultiplier`: multiplication by g maps the space to itself boundedly for the cost gauge) and g *is in the selfs class* (`PointwiseSelfsClass`: the products of g with all atoms have uniformly bounded Besov-ish norm). Keeping these definitions abstract means that the implication "multiplier ⇒ selfs class", which holds for any atom family, lives here, while everything genuinely good-grid-ish is deferred to `GoodGrid/Multipliers/`.

File: `Sums.lean`
Imports (within the project): `none`

Reusable indexing types and notation for sums over grid blocks: the global pair type `BlockIndex` for (k, Q), the types `ChildIndex` and `ParentIndex` for strict descendants and ancestors of a cell, and the corresponding `tsum` notations (`sum^b`, `sum^+`, `sum^-` in the source file).

II. THE GOOD-GRID LIBRARY

`BesovSpacesGoodGrid/GoodGrid/`

In Part II we suppose that $s > 0$, $p \in [1, \infty)$ and $q \in [1, \infty]$; from Section 18 on (the multiplier theory) we further assume $0 < s < 1/p$.

11. Good grids

File: `GoodGrid/Definition.lean`
Imports (within the project): `none (external: UnbalancedHaarWavelet)`

A good grid adds to a weak grid the geometry of a nested dyadic-like structure: levels partition the space (G_4–G_5), each cell sits inside a parent at the previous level (G_6), and — the quantitative heart — the measure of a child is comparable to the measure of its parent, with ratio pinched between λ_1 and λ_2 (G_7). Two consequences are used constantly. First, cell measures decay geometrically in the level, $|Q| \le \lambda_2^k |I|$ for $Q \in \mathcal{P}^k$, so the hypothesis G_2 of Part I holds *automatically* for every $s > 0$: the entire weak-grid theory applies to a good grid with no extra summability assumptions. Second, a cell has at most $1/\lambda_1$ children, which bounds all the branching constants in the Haar construction of Section 15. The simplest example is the dyadic grid of $[0, 1)$ with $\lambda_1 = \lambda_2 = 1/2$; a good grid, however, need not be self-similar, and the two-sided ratio bound is the only regularity retained.

A (λ_1, λ_2)-**good grid**, with $0 < \lambda_1 \le \lambda_2 < 1$, is a grid $\mathcal{P} = (\mathcal{P}^k)_{k\in\mathbb{N}}$ with the following properties:

- G_3. $\mathcal{P}^0 = \{I\}$.
- G_4. $I = \cup_{Q\in\mathcal{P}^k} Q$ up to a set of zero m-measure.
- G_5. The elements of $\mathcal{P}^k$ are pairwise disjoint.
- G_6. For every $Q \in \mathcal{P}^k$, $k > 0$, there exists $P \in \mathcal{P}^{k-1}$ such that $Q \subset P$.
- G_7. We have
$$\lambda_1 \le \frac{|Q|}{|P|} \le \lambda_2$$
for every $Q \subset P$ with $Q \in \mathcal{P}^{k+1}$ and $P \in \mathcal{P}^k$, $k \ge 0$.
- G_8. The family $\cup_k \mathcal{P}^k$ generates the σ-algebra $\mathbb{A}$.

Formalization. Good grids are the structure `GoodGrid` in `GoodGrid/Definition.lean`, which extends the structure `UnbalancedHaarWavelet.Grid` of the external repository (where the nested-partition properties G_3–G_6, G_8 live) with the ratio bounds G_7 (fields `ratio_lower`, `ratio_upper`). The ambient object used by the good-grid files is `GoodGridSpace`, a measurable space with a fixed good grid; the definition `GoodGridSpace.toWeakGrid` turns a good grid into a weak grid (since good-grid partitions are disjoint, the same-level overlap multiplicity is 1), so that the whole of Part I applies. The paper requires $\lambda_1 < \lambda_2$; the formalization allows $\lambda_1 \le \lambda_2$.

```
structure GoodGrid extends UnbalancedHaarWavelet.Grid (α := α) where
        lambda1 : ℝ
        /-- Upper ratio bound (strictly less than 1). -/
        lambda2 : ℝ
```

```
    /-- ‘λ₁‘ is strictly positive. -/
    hlambda1_pos : 0 < lambda1
    /-- ‘λ₂‘ is strictly less than ‘1‘. -/
    hlambda2_lt_one : lambda2 < 1
    /-- ‘λ₁ ≤ λ₂‘. -/
    hlambda1_le_lambda2 : lambda1 ≤ lambda2
    /-- Each child cell’s measure is at least ‘λ₁‘ times its parent’s
measure. -/
    ratio_lower : ∀ (n : ℕ) (s t : Set α),
                    s ∈ grid.partitions (n + 1) → t ∈
grid.partitions n → s ⊆ t →
                    ENNReal.ofReal lambda1 * μ t ≤ μ s
    /-- Each child cell’s measure is at most ‘λ₂‘ times its parent’s
measure. -/
    ratio_upper : ∀ (n : ℕ) (s t : Set α),
                    s ∈ grid.partitions (n + 1) → t ∈
grid.partitions n → s ⊆ t →
                    μ s ≤ ENNReal.ofReal lambda2 * μ t
```

12. Souza's atoms

File: `GoodGrid/BesovSpace.lean`
Imports (within the project): `WeakGrid/Atoms.lean`, `WeakGrid/BesovishSpaces.lean`, `WeakGrid/Completeness.lean`, `WeakGrid/InducedGrid.lean`, `GoodGrid/Definition.lean`

Souza's atoms (paper Section 11A) are the simplest conceivable atoms: a multiple of the indicator of a cell, normalized in L^∞. Their local model space is one-dimensional ($\mathcal{B}(Q) \cong \mathbb{C}$), so the atom set of every cell is a closed bounded subset of a one-dimensional space, hence compact: assumption A_5 comes for free (this is the finite-dimensional criterion of paper Corollary 6.7), and the whole compactness/completeness theory of Part I applies. Their simplicity is the point of the construction: products, restrictions, and positivity of Souza's atoms are trivial to control, which is what makes the multiplier theory of Section 18 tractable.

Let $Q \in \mathcal{P}$. A (s,p)**-Souza's atom** supported on Q is a function $a\colon I \to \mathbb{C}$ such that $a(x) = 0$ for every $x \notin Q$ and a is constant on Q, with

$$|a|_\infty \leq |Q|^{s-1/p}.$$

The set of Souza's atoms supported on Q will be denoted by $\mathcal{A}^{sz}_{s,p}(Q)$. The **canonical Souza's atom** on Q is the Souza's atom with $a(x) = |Q|^{s-1/p}$ for every $x \in Q$. Souza's atoms are (s,p,∞)-type atoms and satisfy A_1–A_5.

Formalization. In `GoodGrid/BesovSpace.lean`. The predicate `IsSouzaAtom` says that a function is supported on a good-grid cell, constant on it, and of size at most $|Q|^{s-1/p}$; the canonical atom is `canonicalSouzaAtom`, with `canonicalSouzaAtom_isSouzaAtom` proving that it is indeed a Souza's atom. The definitions `souzaLocalVectorSpace`, `souzaAtomsSet` and `souzaAtomFamily` package Souza's atoms as an `AtomFamily` with $u = \infty$ satisfying A_1–A_5. The paper's further examples — Hölder atoms (Section 11B) and bounded variation atoms (Section 11C) — are not formalized and are omitted.

```
def IsSouzaAtom (G : GoodGridSpace (α := α)) (s : ℝ) (p : ℝ≥0∞)
    (Q : GoodGridCell G) (a : α → ℂ) : Prop :=
  (∀ x ∉ Q.cell, a x = 0) ∧
  ∃ c : ℂ, (∀ x ∈ Q.cell, a x = c) ∧
          ‖c‖ ≤ (G.grid.μ Q.cell).toReal ^ (s - (p.toReal)⁻¹)
```

13. Besov spaces in a measure space with a good grid

File: `GoodGrid/BesovSpace.lean`
Imports (within the project): `(same file as the previous section)`

The paper's Section 12 introduces the spaces $\mathcal{B}^s_{p,q} = \mathcal{B}^s_{p,q}(\mathcal{P}, \mathcal{A}^{sz}_{s,p})$ and observes that the general theory applies; it states no numbered theorem there. The theorem below is the formalization's packaging of the resulting Souza-space facts, obtained by instantiating Proposition 5.1 and Corollary 6.2 (paper Proposition 6.1 and Corollaries 6.5 and 6.7). The density statements in item C deserve a remark: a function constant on the cells of level k is, by definition, a finite sum of level-k Souza's atoms, and as $k \to \infty$ such "grid-simple" functions approximate any L^p function (this is where G_8, the generation of the σ-algebra by the cells, enters); for $s < 1/p$ the cost of these approximations stays finite, giving density — so, in contrast with classical Besov spaces of positive smoothness on $\mathbb{R}^n$, here the space is a *dense* compactly embedded subspace of L^p.

We study the Besov-ish spaces $\mathcal{B}^s_{p,q} = \mathcal{B}^s_{p,q}(\mathcal{P}, \mathcal{A}^{sz}_{s,p})$ associated with a measure space with a good grid $(I, \mathcal{P}, m)$. If $p \in [1, \infty)$, $q \in [1, \infty]$ and $0 < s < \frac{1}{p}$ we say that $\mathcal{B}^s_{p,q}$ is a **Besov space**. The general theory of Part I specializes as follows.

Theorem 13.1. *Let $(I, \mathcal{P}, m)$ be a measure space with a good grid, $s > 0$, $p \in [1, \infty)$, $q \in [1, \infty]$. Then*

A. *$(\mathcal{B}^s_{p,q}, |\cdot|_{\mathcal{B}^s_{p,q}})$ is a complex Banach space.*
B. *Closed balls of $\mathcal{B}^s_{p,q}$ are compact subsets of L^p, and also of L^1.*
C. *If $0 < s < 1/p$ then $\mathcal{B}^s_{p,q}$ is dense in L^p, and dense in L^1.*

Formalization. All in `GoodGrid/BesovSpace.lean`. Item A is `souzaBesovSpace_costNorm_completeSpace`, the specialization of the weak-grid completeness theorem to Souza's atoms on a good grid. Item B is `souza_closedCostBallInLp_isSeqCompact`, `souza_closedCostBallInLp_isCompact` and `souza_closedCostBallInL1_isCompact`: compactness of closed cost balls in the ambient L^p topology and, after the finite-measure inclusion, in L^1 — the Souza-atom instance of Corollary 6.2 together with the embedding of Proposition 5.1. Item C is `souzaBesovSpace_dense` and `souzaBesovSpace_dense_inL1`, density of the Souza Besov space in L^p and of its image in L^1; the density statements do not appear explicitly as theorems in the paper, and are recorded here because they are formalized.

```
theorem souzaBesovSpace_costNorm_completeSpace
    (G : GoodGridSpace (α := α))
    (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)] :
    @CompleteSpace
      (SouzaBesovSpace G s p q hs hp hp_top)
      (WeakGridSpace.BesovishSpace.costNormedAddCommGroup
        (A := souzaAtomFamily G s p hs hp hp_top) (q := q)
        hp_top (souza_hCco G s p q hs hp
     hp_top)).toMetricSpace.toPseudoMetricSpace.toUniformSpace := ...

theorem souza_closedCostBallInLp_isCompact
    (G : GoodGridSpace (α := α))
    (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    {C : ℝ} (hC : 0 ≤ C) :
    IsCompact
      {f : MeasureTheory.Lp ℂ p G.toWeakGridSpace.measure |
        ∃ hf : f ∈ SouzaBesovSpace G s p q hs hp hp_top,
          WeakGridSpace.BesovishSpace.Norm_Costpq
            (souzaAtomFamily G s p hs hp hp_top) q ⟨f, hf⟩ ≤ C} := ...

theorem souzaBesovSpace_dense
    (G : GoodGridSpace (α := α))
    (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)] :
    Dense (SouzaBesovSpace G s p q hs hp hp_top :
      Set (MeasureTheory.Lp ℂ p G.toWeakGridSpace.measure)) := ...
```

14. Besov's atoms and the Souza–Besov comparison

File: `GoodGrid/BesovAtoms.lean`
Imports (within the project): `GoodGrid/BesovSpace.lean`, `WeakGrid/InducedGrid.lean`, `WeakGrid/Transmutation.lean`

This is the paper's “Alternative characterizations, II: messing with atoms” (Section 16 of the published version), and the first serious application of the transmutation machinery. A Besov atom on Q is not required to be constant

— it is any function supported on Q whose *induced* Besov norm of a higher smoothness $\beta > s$ is controlled by the scale $|Q|^{s-\beta}$; the induced spaces of Section 8 are exactly what makes this definition possible. The comparison theorem (paper Proposition 16.1) says that any atom family sandwiched between Souza's atoms and Besov's atoms generates the *same* space with equivalent norms. The nontrivial inclusion is proved by transmutation: each Besov atom is expanded, inside its own cell, into Souza's atoms with geometrically decaying levelwise mass (rate $\lambda_2^{(k-j)(\beta-s)}$ — this is where $\beta > s$ and the good-grid ratio bound enter), which is precisely hypothesis III of paper Proposition 8.1 with the identity level selector; Claim C of that proposition then converts every representation by Besov atoms into one by Souza's atoms with controlled cost. In practice this proposition is the standard tool for checking that a concrete function belongs to $\mathcal{B}^s_{p,q}$: it suffices to exhibit atoms in any sandwiched class.

Let $s < \beta$ and $\tilde{q} \in [1, \infty]$. A $(s, \beta, p, \tilde{q})$-**Besov atom** on Q is a function $a \in \mathcal{B}^{\beta}_{p,\tilde{q}}(Q, \mathcal{P}_Q, \mathcal{A}^{sz}_{\beta,p})$ such that $a(x) = 0$ for $x \notin Q$ and

$$|a|_{\mathcal{B}^{\beta}_{p,\tilde{q}}(Q,\mathcal{P}_Q,\mathcal{A}^{sz}_{\beta,p})} \leq \frac{1}{C_7}|Q|^{s-\beta},$$

where $C_7 = C_7(C_1, \lambda_2, \beta, \tilde{q}, p) \geq 1$ is the normalizing constant of the paper. The family of $(s, \beta, p, \tilde{q})$-Besov atoms supported on Q is denoted $\mathcal{A}^{bs}_{s,\beta,p,\tilde{q}}(Q)$, and $\mathcal{A}^{sz}_{s,p}(Q) \subset \mathcal{A}^{bs}_{s,\beta,p,\tilde{q}}(Q)$; Besov atoms are atoms of type $(s, p, 1)$.

Proposition 14.1 (Souza's atoms and Besov's atoms)**.** *Let $\mathcal{P}$ be a good grid and $0 < s < 1/p$. Let $\mathcal{A}$ be a class of (s, p, u)-atoms, with $u \geq 1$, such that for some $s < \beta$, $\tilde{q} \in [1, \infty]$, and $C_8, C_9 \geq 0$ we have, for every $Q \in \mathcal{P}$,*

$$\frac{1}{C_8}\mathcal{A}^{sz}_{s,p}(Q) \subset \mathcal{A}(Q) \subset C_9\, \mathcal{A}^{bs}_{s,\beta,p,\tilde{q}}(Q).$$

Then

$$\mathcal{B}^s_{p,q}(\mathcal{P}, \mathcal{A}^{sz}_{s,p}) = \mathcal{B}^s_{p,q}(\mathcal{P}, \mathcal{A}) = \mathcal{B}^s_{p,q}(\mathcal{P}, \mathcal{A}^{bs}_{s,\beta,p,\tilde{q}}),$$

with

$$|f|_{\mathcal{B}^s_{p,q}(\mathcal{A})} \leq C_8\, |f|_{\mathcal{B}^s_{p,q}(\mathcal{A}^{sz}_{s,p})} \quad \textit{and} \quad |f|_{\mathcal{B}^s_{p,q}(\mathcal{A}^{sz}_{s,p})} \leq C\, |f|_{\mathcal{B}^s_{p,q}(\mathcal{A})}.$$

Formalization. This is paper Proposition 16.1. The sandwich proposition is the theorem `atoms_between_souza_atoms_and_besov_atoms` in `GoodGrid/BesovAtoms.lean`: it proves the equality of the three Besov-ish spaces and both continuous norm bounds, the second with the explicit constant $C_2/(1 - \lambda_2^{\beta-s})$. It is supported by the decay claims `besovAtom_to_souza_representation_decay` and `besovAtom_to_induced_souzaS_representation_decay_claimC` (the Lean counterparts of the Claim in the paper's proof: a single Besov atom, read on the weak grid induced on its supporting cell, admits a Souza-atom representation with geometric coefficient decay rate $\lambda_2^{(k-j)(\beta-s)}$), by the normalizing constant `besovAtomConstant` (with `one_le_besovAtomConstant` giving the lower bound

$1 \le C_7$ used to simplify the final constant), and by the identity-level geometric-sum computation `LpGridRepresentation.cCoefficientInt_transmutationKernelZ_zero_one` of Section 9. Paper Propositions 16.2 (Hölder atoms) and 16.3 (bounded variation atoms) are not formalized and are omitted.

```
theorem atoms_between_souza_atoms_and_besov_atoms
    (G : GoodGridSpace (α := α)) (s β : ℝ) (p u q qtilde : ℝ≥0∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ u)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)]
    (hs : 0 < s) (hβ : 0 < β) (hβs : s < β) (hp_top : p ≠ ∞)
    (A : WeakGridSpace.AtomFamily G.toWeakGridSpace s p u)
    (C1 C2 : ℝ)
    (hSandwich :
      SouzaBesovSandwich G s β p u qtilde hs hβ (inferInstance : Fact (1 ≤ p))
        hp_top A C1 C2) :
    (WeakGridSpace.BesovishSpace (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p)
     hp_top) q =
        WeakGridSpace.BesovishSpace A q) ∧
      (WeakGridSpace.BesovishSpace A q =
        WeakGridSpace.BesovishSpace
          (besovAtomFamily G s β p qtilde hs hβ inferInstance hp_top) q) ∧
      (∀ f : WeakGridSpace.BesovishSpace
          (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q,
        ∃ hfA : WeakGridSpace.MemBesovishCoeffCost A q
            (f : Lp ℂ p G.toWeakGridSpace.measure),
          WeakGridSpace.BesovishSpace.Norm_Costpq A q
              (⟨(f : Lp ℂ p G.toWeakGridSpace.measure), hfA⟩ :
                WeakGridSpace.BesovishSpace A q)
            ≤ C1 *
              WeakGridSpace.BesovishSpace.Norm_Costpq
                (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q f) ∧
      (∀ f : WeakGridSpace.BesovishSpace A q,
        ∃ hfS : WeakGridSpace.MemBesovishCoeffCost
            (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q
            (f : Lp ℂ p G.toWeakGridSpace.measure),
          WeakGridSpace.BesovishSpace.Norm_Costpq
              (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q
              (⟨(f : Lp ℂ p G.toWeakGridSpace.measure), hfS⟩ :
                WeakGridSpace.BesovishSpace
                  (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q)
            ≤ (C2 / (1 - G.grid.lambda2 ^ (β - s))) *
              WeakGridSpace.BesovishSpace.Norm_Costpq A q f) := ...
```

15. Unbalanced Haar wavelets

File: `GoodGrid/AlternativeRepresentationsAndNorms/HaarRepresentationNorm.lean`
Imports (within the project): `GoodGrid/BesovSpace.lean (external: UnbalancedHaarWavelet, UnconditionalSchauderBasis, Burkholder)`

On a good grid there is a genuine wavelet system: since each cell has between 2 and $1/\lambda_1$ children, one can follow Girardi and Sweldens and organize the children of each cell into a balanced binary tree of subsets (the "logarithmic subtrees"), attaching to each internal node a two-valued zero-mean function — an *unbalanced Haar function*. The measure-ratio bounds G_7 guarantee that each ϕ_S supported in Q has size $\approx |Q|^{-1/2}$ on its support, so the system behaves like the classical Haar basis even though the grid may be very irregular. The unconditionality of this basis in L^β, $1 < \beta < \infty$ — a Burkholder-type martingale inequality — is the one substantial analytic ingredient that the repository does not prove internally: it is supplied by the external formalized libraries listed above.

Let $\mathcal{P} = (\mathcal{P}^k)_k$ be a good grid. Following Girardi and Sweldens, one constructs, from the children structure of the grid (Type I tree with the logarithmic subtrees construction), a family of **unbalanced Haar functions** $\{\phi_S\}_{S\in\hat{\mathcal{H}}}$, where $\hat{\mathcal{H}} = \{I\} \cup \mathcal{H}$, $\phi_I = 1_I/|I|^{1/2}$, each ϕ_S with $S \in \mathcal{H}_Q$ is supported on Q, has zero mean, L^2-norm one, and satisfies

$$\frac{C_5(\lambda_1,\lambda_2)}{|Q|^{1/2}} \le |\phi_S(x)| \le \frac{C_6(\lambda_1,\lambda_2)}{|Q|^{1/2}}$$

on its support. The family $\{\phi_S\}_{S\in\hat{\mathcal{H}}}$ is an unconditional basis of L^β for every $1 < \beta < \infty$.

Formalization. The construction and the unconditional basis property are provided by the external Lean repositories `UnbalancedHaarWavelet`, `UnconditionalSchauderBasis`, `LaminarFamiliesMaximalBinaryTrees` and `Burkholder` (SmaniaD), on which this repository depends. The local API (L^2-normalized Haar functions `L2normalizedHaar` and Haar coefficients `Coeff`) is in `GoodGrid/AlternativeRepresentationsAndNorms/HaarRepresentationNorm.lean`.

```
def L2normalizedHaar (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))
    (i : F.Index) (x : α) : ℂ :=
  ((l2NormalizationFactor G F i : ℝ) : ℂ) *
    (UnbalancedHaarWavelet.FullHaarSystem.function (GridOf G) F i x : ℂ)

def Coeff (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))
    (f : α → ℂ) (_hf : Integrable f G.grid.μ) (i : F.Index) : ℂ :=
  ∫ x, f x * L2normalizedHaar G F i x ∂G.grid.μ
```

16. Alternative norms: standard, Haar, and mean oscillation

File: `GoodGrid/AlternativeRepresentationsAndNorms/` (eleven files, see Section 19)
Imports (within the project): `GoodGrid/BesovSpace.lean`, `WeakGrid/InducedGrid.lean`, `HaarRepresentationNorm.lean`

This is the paper's "Alternative characterizations, I: messing with norms" (Section 15 of the published version). The Besov norm is an infimum over all representations, which makes it hard to compute for a concrete f; this section provides three norms defined directly from f and proves them equivalent to it. The *Haar norm* N_{haar} weighs the Haar coefficients $d^f_S = \int f \phi_S \, dm$; the *standard norm* N_{st} weighs the coefficients of one distinguished atomic representation, obtained by resumming the Haar expansion cellwise (each k^f_P is a finite, geometry-bounded combination of the d^f_S with S in the Haar tree of the parent of P); and the *mean-oscillation norm* N_{osc} measures, in the spirit of Dorronsoro, how well f is approximated by constants at every cell and scale. The equivalences are proved as a cycle $|\cdot|_{\mathcal{B}^s_{p,q}} \to N_{st} \to N_{haar} \to N_{osc} \to |\cdot|_{\mathcal{B}^s_{p,q}}$ (inequalities (15-30)–(15-33) of the paper), each step with its own mechanism: testing $f - c_Q$ against the zero-mean wavelets bounds Haar coefficients by oscillations; the constancy of low-frequency Souza blocks on finer cells means oscillation only sees the high-frequency tail of a representation, which a discrete convolution estimate sums; and the standard representation, being an actual representation, dominates the infimum norm. In particular the standard representation converges to f for *every* f in the space — a canonical, linear-in-f decomposition that substitutes for the nonuniqueness of general representations.

For $f \in L^\beta$, $\beta > 1$, write $d^f_S = \int f \phi_S \, dm$ for the Haar coefficients of f, and define

$$N_{haar}(f) = |I|^{1/p-s-1/2}|d^f_I| + \Big(\sum_k \big(\sum_{Q\in\mathcal{P}^k} |Q|^{1-sp-\frac{p}{2}} \sum_{S\in\mathcal{H}_Q} |d^f_S|^p\big)^{q/p}\Big)^{1/q}.$$

From the Haar coefficients one builds, exactly as in the paper, the **standard atomic representation**

$$f = \sum_i \sum_{Q\in\mathcal{P}^i} k^f_Q \, a_Q$$

(a_Q canonical Souza's atoms) and its norm

$$N_{st}(f) = |k^f_I| + \Big(\sum_{k\geq 1} \big(\sum_{Q\in\mathcal{P}^k} |k^f_Q|^p\big)^{q/p}\Big)^{1/q}.$$

Finally, with

$$osc_p(f, Q) = \inf_{c\in\mathbb{C}} \Big(\int_Q |f - c|^p \, dm \Big)^{1/p},$$

define

$$N_{osc}(f) = |I|^{-s}|f|_p + \Big(\sum_k \big(\sum_{Q\in\mathcal{P}^k} |Q|^{-sp} osc_p(f, Q)^p \big)^{q/p} \Big)^{1/q}.$$

Theorem 16.1 (Equivalence of norms)**.** *Suppose $0 < s < 1/p$, $p \in [1, \infty)$ and $q \in [1, \infty]$. Each one of the quantities $|f|_{\mathcal{B}^s_{p,q}}$, $N_{st}(f)$, $N_{haar}(f)$, $N_{osc}(f)$ is finite if and only if $f \in \mathcal{B}^s_{p,q}$, and these norms are equivalent on $\mathcal{B}^s_{p,q}$. More precisely there are constants, depending only on the grid geometry and on s, p, q, such that*

$$N_{st}(f) \le C\,|f|_{\mathcal{B}^s_{p,q}}, \qquad N_{st}(f) \le C\,N_{haar}(f),$$
$$N_{haar}(f) \le C\,N_{osc}(f), \qquad N_{osc}(f) \le C\,|f|_{\mathcal{B}^s_{p,q}},$$

and conversely: if $N_{st}(f) < \infty$ then $f \in \mathcal{B}^s_{p,q}$ and the standard atomic representation is a $\mathcal{B}^s_{p,q}$-representation of f (so $|f|_{\mathcal{B}^s_{p,q}} \le N_{st}(f)$); if $N_{haar}(f) < \infty$ and $f \in L^1$ then $f \in L^p$ and the Haar series of f converges in L^p.

Formalization. In the files of `GoodGrid/AlternativeRepresentationsAndNorms/` . The equivalences are proved as a cycle of separate inequalities, each with a finite existential constant: `exists_standardRepresentationNorm_le_const_mul_souzaBesovNorm` ($N_{st} \le C|\cdot|_{\mathcal{B}^s_{p,q}}$), `exists_standardRepresentationNorm_le_const_mul_haarL2RepresentationNorm` (the Haar gauge controls the standard atomic gauge), `exists_haarL2RepresentationNorm_le_const_mul_meanOscillationNorm` (mean oscillation controls the Haar gauge; the cellwise estimate is the Lean analogue of testing $f - c_Q$ against the zero-mean Haar wavelets over a parent cell) and `exists_meanOscillationNorm_le_const_mul_souzaBesovNorm` (the Besov norm controls mean oscillation; the proof separates the L^p term from the oscillation seminorm and uses that low-frequency Souza blocks are constant on finer cells, so local oscillation is controlled by the high-frequency tail and a geometric discrete convolution estimate). The converse endpoints are `finite_standardRepresentationNorm_implies_memBesov_and_standardRepresentation` (finite N_{st} gives L^p membership, the canonical standard representation as a genuine finite-cost $\mathcal{B}^s_{p,q}$-representation, and Souza-Besov membership with the expected cost bounds) and `finite_haarL2RepresentationNorm_implies_memLp_and_hasSum` (finite N_{haar} gives $f \in L^p$ and convergence of the normalized Haar expansion to f in L^p); the wrap-up of the Haar-side comparisons is `exists_haarRepresentationNorms_equivalent`. Compared with paper Theorem 15.1: the formalization proves the equivalences as a cycle of separate inequalities with existential constants (the explicit constants θ_1, θ_2 of the paper are not tracked), it works with the L^2-normalized Haar coefficients, and there is no single wrap-up statement packaging all four norms at once.

```lean
theorem exists_standardRepresentationNorm_le_const_mul_haarL2RepresentationNorm
    (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (F : UnbalancedHaarWavelet.FullHaarSystem (G := HaarRepresentation.GridOf G))
    [DecidableEq F.Index]
    (s : ℝ) (hs : 0 < s)
    (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
    (q : ℝ≥0∞) [Fact (1 ≤ q)] :
    ∃ C : ℝ≥0∞, C ≠ ∞ ∧
      ∀ (f : α → ℂ) (hfint : Integrable f G.grid.μ),
        HaarRepresentation.haarL2RepresentationNorm G F s p q f hfint ≠ ∞ →
          standardRepresentationNorm G F s hs p hp_top q f hfint ≠ ∞ ∧
            standardRepresentationNorm G F s hs p hp_top q f hfint ≤
              C * HaarRepresentation.haarL2RepresentationNorm G F s p q f hfint :=
     ...

theorem exists_haarL2RepresentationNorm_le_const_mul_meanOscillationNorm
    (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))
    (s : ℝ) (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
    (q : ℝ≥0∞) [Fact (1 ≤ q)] :
    ∃ C : ℝ≥0∞, C ≠ ∞ ∧
      ∀ (f : α → ℂ) (hf : Integrable f G.grid.μ), MemLp f p G.grid.μ →
        MeanOscillation.meanOscillationNorm G s p q f ≠ ∞ →
          haarL2RepresentationNorm G F s p q f hf ≤
            C * MeanOscillation.meanOscillationNorm G s p q f := ...

theorem exists_meanOscillationNorm_le_const_mul_souzaBesovNorm
    (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (s : ℝ) (hs : 0 < s)
    (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
    (q : ℝ≥0∞) [Fact (1 ≤ q)] :
    ∃ C : ℝ≥0∞, C ≠ ∞ ∧
      ∀ (g : SouzaBesovSpace G s p q hs Fact.out (ne_of_lt hp_top))
        (f : α → ℂ),
          f =ᵐ[G.grid.μ] ((g : Lp ℂ p G.toWeakGridSpace.measure) : α → ℂ) →
            meanOscillationNorm G s p q f ≤
              C * ENNReal.ofReal
                (WeakGridSpace.BesovishSpace.Norm_Costpq
                  (souzaAtomFamily G s p hs Fact.out (ne_of_lt hp_top)) q g) := ...

theorem
     finite_standardRepresentationNorm_implies_memBesov_and_standardRepresentation
    (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (F : UnbalancedHaarWavelet.FullHaarSystem (G := HaarRepresentation.GridOf G))
    [DecidableEq F.Index]
    (s : ℝ) (hs : 0 < s)
    (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
    (q : ℝ≥0∞) [Fact (1 ≤ q)]
    (f : α → ℂ) (hf : Integrable f G.grid.μ)
    (hN : standardRepresentationNorm G F s hs p hp_top q f hf ≠ ∞) :
    ∃ hfLp : MemLp f p G.grid.μ,
      ∃ g : SouzaBesovSpace G s p q hs Fact.out (ne_of_lt hp_top),
        ∃ R :
          WeakGridSpace.LpGridRepresentation
```

```
        (souzaAtomFamily G s p hs Fact.out (ne_of_lt hp_top))
        (g : Lp ℂ p G.toWeakGridSpace.measure),
      (g : Lp ℂ p G.toWeakGridSpace.measure) = hfLp.toLp f ∧
        R.block = canonicalStandardBlockSeq G F s hs p hp_top f hf ∧
          WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R ∧
            WeakGridSpace.LpGridRepresentation.pqCostENNReal (q := q) R =
              standardRepresentationNorm G F s hs p hp_top q f hf ∧
              WeakGridSpace.LpGridRepresentation.pqCost (q := q) R ≤
                (standardRepresentationNorm G F s hs p hp_top q f hf).toReal ∧
                WeakGridSpace.BesovishSpace.Norm_Costpq
                    (souzaAtomFamily G s p hs Fact.out (ne_of_lt hp_top)) q g
≤
                  (standardRepresentationNorm G F s hs p hp_top q f hf).toReal
:= ...
```

Corollary 16.2 (Reading the standard coefficients)**.** *For each $P \in \mathcal{P}$ the coefficient k^f_P of the standard atomic representation of f is a (finite) linear combination of the Haar coefficients d^f_S, $S \in \mathcal{H}_Q$, of f, and the standard representation of any $f \in \mathcal{B}^s_{p,q}$ converges to f in L^p with*

$$\Big(\sum_k \Big(\sum_{P\in\mathcal{P}^k} |k^f_P|^p\Big)^{q/p}\Big)^{1/q} \le C\,|f|_{\mathcal{B}^s_{p,q}}.$$

Formalization. In `GoodGrid/AlternativeRepresentationsAndNorms/HaarRepresentationNorm.lean`, together with the theorems of the previous *Formalization* paragraph. The Haar coefficients are the definition `Coeff`, and the theorem `hasSum_coeff_smul_l2normalizedHaar_toLp` proves that the normalized Haar expansion of any $f \in L^\beta$, $1 < \beta < \infty$, converges to f. Paper Corollary 15.2 states moreover that $f \mapsto k^f_P$ extends to a bounded linear functional on L^1; this extension is not formalized.

```
theorem hasSum_coeff_smul_l2normalizedHaar_toLp
    (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
    (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))
    [DecidableEq F.Index]
    (β : ℝ≥0∞) (hβ_one : 1 < β) (hβ_top : β < ∞)
    (f : α → ℂ) (hf : MemLp f β G.grid.μ) :
    letI : Fact (1 ≤ β) := ⟨le_of_lt hβ_one⟩
    HasSum
      (fun i : F.Index =>
        Coeff G F f
            (by
              letI : IsFiniteMeasure G.grid.μ := (GridOf G).isFinite
              exact hf.integrable (le_of_lt hβ_one))
            i •
          (l2normalizedHaar_memLp G F β i).toLp (L2normalizedHaar G F i))
      (hf.toLp f) := ...
```

17. Positive cone

File: `GoodGrid/PositiveCone.lean`
Imports (within the project): `GoodGrid/BesovSpace.lean`, `GoodGrid/AlternativeRepresentationsAndNorms/StandarRepresentationNormleqBesovNorm.lean`, `WeakGrid/Multipliers.lean`

Positivity is where Souza's atoms shine: since canonical atoms are *positive multiples of indicators*, a representation with nonnegative coefficients is a sum of nonnegative functions, and no cancellation can occur. This rigidity has strong structural consequences, proved in this file. The support of a positive function is (mod m) a countable union of cells, and — a sharpening obtained in the formalization — in *any* positive representation of a function supported in a cell, all active cells must sit inside that cell: a coefficient cannot be "wasted" outside the support, because nothing can cancel it. This support rigidity is the engine of the positive-cone non-Archimedean theorems of Section 18. The original motivation comes from dynamics: the fixed points of transfer operators of expanding maps — invariant densities — are nonnegative and often discontinuous, and the positive cone is their natural home inside the Besov space. The file depends on the standard-representation comparison of Section 16 (hence its position in the import order): the decomposition $f = f_+ - f_-$ is obtained by splitting the coefficients of the *standard* representation, whose linearity and canonicity make the positive parts well defined.

We say that f is $\mathcal{B}^s_{p,q}$-positive if there is a $\mathcal{B}^s_{p,q}$-representation

$$f = \sum_k \sum_{P \in \mathcal{P}^k} c_P a_P$$

where $c_P \geq 0$ and a_P is the canonical (s,p)-Souza's atom supported on P. The set of all $\mathcal{B}^s_{p,q}$-positive functions is a convex cone in $\mathcal{B}^s_{p,q}$, denoted $\mathcal{B}^{s+}_{p,q}$, with the "norm"

$$|f|_{\mathcal{B}^{s+}_{p,q}} = \inf \Big(\sum_k \big(\sum_{P \in \mathcal{P}^k} c_P^p \big)^{q/p} \Big)^{1/q},$$

the infimum running over all positive representations of f.

Formalization. In `GoodGrid/PositiveCone.lean`. A positive representation (canonical atoms, nonnegative coefficients) is the predicate `SouzaPositiveRepresentation`, and the cone of $\mathcal{B}^s_{p,q}$-positive functions is `souzaPositiveCone`, with the positive "norm" `souzaPositiveNorm`. The formalization also uses the wider notion of a *cone-positive* representation (`SouzaConePositiveRepresentation`): real coefficients ≥ 0 and Souza's atoms which are a.e. nonnegative real-valued, not necessarily canonical. The file also provides the additive infrastructure used in the non-Archimedean theorems of Section 18: sums and scalar multiples of positive representations (`souzaPositiveRepresentationAdd*`, `exists_souzaPositiveRepresentation_finset_sum`) with levelwise Minkowski control of the cost (`souzaPositiveRepresentationAdd_levelCoeffRoot_le`).

```
def SouzaPositiveRepresentation
    (G : GoodGridSpace (α := α)) (s : ℝ) (p : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)]
    {f : Lp ℂ p G.toWeakGridSpace.measure}
    (R : WeakGridSpace.LpGridRepresentation
      (souzaAtomFamily G s p hs hp hp_top) f) : Prop :=
  ∀ k, SouzaPositiveLevelBlock G s p hs hp hp_top (R.block k)

def SouzaConePositiveRepresentation
    (G : GoodGridSpace (α := α)) (s : ℝ) (p : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)]
    {f : Lp ℂ p G.toWeakGridSpace.measure}
    (R : WeakGridSpace.LpGridRepresentation
      (souzaAtomFamily G s p hs hp hp_top) f) : Prop :=
  ∀ k, SouzaConePositiveLevelBlock G s p hs hp hp_top (R.block k)
```

Proposition 17.1. *Let $0 < s < 1/p$.*

A. *If $f \in \mathcal{B}^s_{p,q}$ is (a.e.) real valued then one can find $f_+, f_- \in \mathcal{B}^{s+}_{p,q}$ such that*
$$f = f_+ - f_-,$$
with the positive costs of f_+ and f_- controlled in terms of $|f|_{\mathcal{B}^s_{p,q}}$; an analogous four-term decomposition holds for complex valued f.

B. *The positive cone $\mathcal{B}^{s+}_{p,q}$ is dense in the cone of nonnegative functions of L^p.*

Formalization. In `GoodGrid/PositiveCone.lean`. Item A is `exists_souzaPositive_decomposition_of_aeRealValued` (the decomposition $f = f_+ - f_-$ with controlled positive costs, for a.e. real valued f) and `exists_souzaPositive_decomposition_of_aeComplexValued` (the four-term decomposition for complex valued f); in the paper this appears as an unnumbered observation in Section 13, and the complex case is not stated there. Item B is `souzaPositiveCone_dense_in_LpNonnegativeCone`, density of the positive cone in the nonnegative cone of L^p; not stated in the paper.

```
theorem exists_souzaPositive_decomposition_of_aeRealValued
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)] :
    ∃ C : ℝ,
```

```
      0 ≤ C ∧
        ∀ f : WeakGridSpace.BesovishSpace
            (souzaAtomFamily G s p hs hp hp_top) q,
          SouzaBesovAEEqRealValued G s p q hs hp hp_top f →
            ∃ u v : WeakGridSpace.BesovishSpace
                (souzaAtomFamily G s p hs hp hp_top) q,
              SouzaPositiveElement G s p q hs hp hp_top u ∧
                SouzaPositiveElement G s p q hs hp hp_top v ∧
                f = u - v ∧
                souzaPositiveNorm G s p q hs hp hp_top u ≤
                  ENNReal.ofReal C *
                    ENNReal.ofReal (WeakGridSpace.BesovishSpace.Norm_Costpq
                      (souzaAtomFamily G s p hs hp hp_top) q f) ∧
                souzaPositiveNorm G s p q hs hp hp_top v ≤
                  ENNReal.ofReal C *
                    ENNReal.ofReal (WeakGridSpace.BesovishSpace.Norm_Costpq
                      (souzaAtomFamily G s p hs hp hp_top) q f) := ...

theorem souzaPositiveCone_dense_in_LpNonnegativeCone
    (G : GoodGridSpace (α := α)) (s : ℝ) (β q : ℝ≥0∞)
    (hs : 0 < s) (hβ : 1 ≤ β) (hβ_top : β ≠ ∞)
    [Fact (1 ≤ β)] [Fact (1 ≤ q)] :
    LpNonnegativeCone G β ⊆
      closure (SouzaPositiveConeInLbeta G s β q hs hβ hβ_top) := ...
```

Proposition 17.2 (Support of positive functions). *If $f \in \mathcal{B}^{s+}_{p,q}$ then its support*

$$\operatorname{supp} f = \{x \in I\colon\ f(x) \neq 0\}$$

is, up to a set of zero measure, a countable union of elements of $\mathcal{P}$. Moreover, in any positive representation of f, the coefficient of a cell Q vanishes unless Q is contained (mod m) in this union.

Formalization. In `GoodGrid/PositiveCone.lean` (paper Proposition 13.1). The first sentence is `support_ae_countable_iUnion_goodGridCells_of_souzaPositiveFunction`, which exhibits the support of a positive function as an a.e. countable union of good-grid cells. The second sentence is the support theorem `souzaPositiveRepresentation_coeff_eq_zero_of_not_subset_cell`, a sharper statement obtained in the formalization: in any positive representation of a function supported in a cell, every active cell is contained in that cell. A companion "linchpin" lemma, `souzaPositiveRepresentation_ae_ne_zero_on_active_cell`, proves that a positively represented function is a.e. nonzero on every active cell of its representation; both are used repeatedly in the positive-cone non-Archimedean arguments of Section 18.

```
theorem support_ae_countable_iUnion_goodGridCells_of_souzaPositiveFunction
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    {f : α → ℂ}
```

```
    (hf : SouzaPositiveFunction G s p q hs hp hp_top f) :
    IsAECountableUnionOfGoodGridCells G {x | f x ≠ 0} := ...

theorem souzaPositiveRepresentation_coeff_eq_zero_of_not_subset_cell
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    (Qc : GoodGridCell G)
    {g : Lp ℂ p G.toWeakGridSpace.measure}
    (R : WeakGridSpace.LpGridRepresentation (souzaAtomFamily G s p hs hp
    hp_top) g)
    (hR : SouzaPositiveRepresentation G s p hs hp hp_top R)
    (hsupp : ∀ᵐ x ∂ G.toWeakGridSpace.measure, x ∉ Qc.cell → (g : α → ℂ
    ) x = 0)
    {n : ℕ} (P : WeakGridSpace.LevelCell G.toWeakGridSpace n)
    (hP : ¬ P.1 ⊆ Qc.cell) :
    (R.block n).coeff P = 0 := ...
```

18. Pointwise multipliers acting on $\mathcal{B}^s_{p,q}$

This section documents the directory `GoodGrid/Multipliers/` (five files plus the standalone interface), which formalizes Section 18 of the published paper. From here on $0 < s < 1/p$. The guiding question is: for which g is $f \mapsto gf$ bounded on $\mathcal{B}^s_{p,q}$? Because the space is built from Souza's atoms, and the product of g with a Souza's atom is just g restricted to a cell and rescaled, the natural test is to multiply g against every atom — this gives the *selfs* classes, and the theory consists in showing that this obviously necessary condition is, in various quantified senses, also sufficient. The subsections below follow the import order of the files.

18.1. The selfs classes and their seminorms.

File: `GoodGrid/Multipliers/Definition.lean`
Imports (within the project): `WeakGrid/Multipliers.lean`, `GoodGrid/BesovSpace.lean`

Let $g\colon I \to \mathbb{C}$ be a measurable function. We say that g is a **pointwise multiplier** acting on $\mathcal{B}^s_{p,q}$ if the transformation $G(f) = gf$ defines a bounded operator in $\mathcal{B}^s_{p,q}$. We denote the set of pointwise multipliers by $M(\mathcal{B}^s_{p,q})$, with the norm

$$|g|_{M(\mathcal{B}^s_{p,q})} = \sup\{|gf|_{\mathcal{B}^s_{p,q}}\colon\ |f|_{\mathcal{B}^s_{p,q}} \leq 1\}.$$

Define

$$\mathcal{B}^s_{p,q,selfs} = \Big\{g \in \mathcal{B}^s_{p,q}\colon \sup_{a_Q \in \mathcal{A}^{sz}_{s,p}} |ga_Q|_{\mathcal{B}^s_{p,q}} < \infty\Big\},$$

with the norm

$$|g|_{\mathcal{B}^s_{p,q,selfs}} = \sup_{a_Q \in \mathcal{A}^{sz}_{s,p}} |ga_Q|_{\mathcal{B}^s_{p,q}},$$

and, for $t \in \mathbb{N}$, the tail seminorms

$$|g|_{\mathcal{B}^{s,t}_{p,q,selfs}} = \sup_{\substack{a_Q \in \mathcal{A}^{sz}_{s,p} \\ Q \in \mathcal{P}^j,\ j \geq t}} |g a_Q|_{\mathcal{B}^s_{p,q}}.$$

Formalization. In GoodGrid/Multipliers/Definition.lean. Pointwise multipliers are the predicate SouzaPointwiseMultiplier, the class $\mathcal{B}^s_{p,q,selfs}$ is SouzaPointwiseSelfsClass, and the tail variant is SouzaPointwiseSelfsTailClass: the quantitative form SouzaPointwiseSelfsTailBound g C records that for every Souza's atom on a cell of level at least t the product $g\,a_Q$ admits a representation of cost at most C, and the induced seminorm $|\cdot|_{\mathcal{B}^{s,t}_{p,q,selfs}}$ is souzaPointwiseSelfsTailNorm.

```
abbrev SouzaPointwiseMultiplier
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    (m : α → ℂ) : Prop :=
  WeakGridSpace.IsPointwiseMultiplier
    (A := souzaAtomFamily G s p hs hp hp_top) q m

abbrev SouzaPointwiseSelfsClass
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    (m : α → ℂ) : Prop :=
  WeakGridSpace.PointwiseSelfsClass
    (A := souzaAtomFamily G s p hs hp hp_top) q m
```

18.2. **Restriction to a cell.**

File: GoodGrid/Multipliers/Besovspq.lean
Imports (within the project): GoodGrid/Multipliers/Definition.lean, WeakGrid/Transmutation.lean

The first multiplier theorem is for the simplest candidates: indicators of cells. The proof is another instance of transmutation — restricting a Souza's atom a_Q to a cell W produces either 0, the atom itself, or (when $W \subset Q$) a rescaled atom on W, with geometric decay in the level gap; Proposition 8.1(A) and (B) of the paper then assemble these local expansions with uniform cost. The lemma is stated on the induced grid of W, and the uniformity of C_{22} in W is what the non-Archimedean arguments below need.

Lemma 18.1 (Restriction to a cell)**.** *Let $W \in \mathcal{P}$. The restriction map*

$$r\colon \mathcal{B}^s_{p,q}(I, \mathcal{P}, \mathcal{A}^{sz}_{s,p}) \to \mathcal{B}^s_{p,q}(W, \mathcal{P}_W, \mathcal{A}^{sz}_{s,p}), \qquad r(f) = 1_W f,$$

is continuous: there is $C_{10} \geq 1$, independent of W, such that

A. *For every* $f \in \mathcal{B}^s_{p,q}$ *we have*

$$|1_W f|_{\mathcal{B}^s_{p,q}(W,\mathcal{P}_W,\mathcal{A}^{sz}_{s,p})} \le C_{10}\, |f|_{\mathcal{B}^s_{p,q}}.$$

In particular $1_W \in M(\mathcal{B}^s_{p,q})$ *with* $|1_W|_{M(\mathcal{B}^s_{p,q})} \le C_{10}$, *uniformly in* W.

B. *For every* $\mathcal{B}^{s+}_{p,q}$*-representation* $f = \sum_k \sum_Q c_Q a_Q$ *one can find a positive representation*

$$1_W f = \sum_k \sum_Q d_Q\, a_Q$$

on the induced grid with cost at most C_{10} *times the cost of the original one, and such that* $d_Q \neq 0$ *implies* $Q \subset \operatorname{supp} f$ *(mod* m*).*

Formalization. In `GoodGrid/Multipliers/Besovspq.lean` (paper Lemma 18.2). The restriction map with its induced-grid representation is `souzaCellIndicatorRestrictsToInduced`; the cost bounds (uniform in W) are the lemmas `souzaIndicatorRestrictionBound_of_ambientRestrictionTransmutation_*` and `souzaCellIndicatorPointwiseMultiplierBound_uniform`, the latter giving the explicit multiplier bound $2C + 1$ with C = `souzaAmbientRestrictionMultiplierConstant`; and the theorem `souzaCellIndicatorPointwiseMultiplier` concludes that 1_W is a pointwise multiplier of $\mathcal{B}^s_{p,q}$. The paper's strongly regular domains (Definition 18.7 and Proposition 18.8), of which the present lemma is the grid-cell case, are not formalized in general; the multiplier statement for 1_W, $W \in \mathcal{P}$, with a bound uniform in W, is the formalized special case.

```
theorem souzaCellIndicatorPointwiseMultiplier
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    (hs_lt_inv : s < (p.toReal)⁻¹)
    (W : GoodGridCell G) :
    SouzaPointwiseMultiplier G s p q hs hp hp_top
      (W.cell.indicator fun _ => (1 : ℂ)) := ...

theorem souzaCellIndicatorPointwiseMultiplierBound_uniform
    (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    (hs_lt_inv : s < (p.toReal)⁻¹)
    (W : GoodGridCell G) :
    SouzaPointwiseMultiplierBound G s p q hs hp hp_top
      (W.cell.indicator fun _ => (1 : ℂ))
      (2 * souzaAmbientRestrictionMultiplierConstant G s p + 1) := ...
```

18.3. The multipliers of $\mathcal{B}^s_{1,1}$.

File: `GoodGrid/Multipliers/Besovs11.lean`
Imports (within the project): `GoodGrid/Multipliers/Definition.lean`

For $p = q = 1$ the multiplier problem has a complete and clean answer: the necessary condition is sufficient. The reason is that for an ℓ^1-type cost one can substitute the representation of f into gf atom by atom, $gf = \sum c_Q(ga_Q)$, and the costs of the pieces ga_Q simply add up — no interaction between cells survives. This is paper Proposition 18.1.

Proposition 18.2. *We have that*

$$M(\mathcal{B}^s_{1,1}) = \mathcal{B}^s_{1,1,selfs}.$$

Formalization. This is paper Proposition 18.1. The equivalence is the theorem `souzaPointwiseMultiplier_iff_souzaPointwiseSelfsClass_one_one`: the inclusion $\mathcal{B}^s_{1,1,selfs} \subset M(\mathcal{B}^s_{1,1})$ is proved in `GoodGrid/Multipliers/Besovs11.lean`, while the converse, which holds for all p, q, is `souzaPointwiseSelfsClass_of_souzaPointwiseMultiplier` in `GoodGrid/Multipliers/Besovspq.lean`.

```
theorem souzaPointwiseMultiplier_iff_souzaPointwiseSelfsClass_one_one
    (G : GoodGridSpace (α := α)) (s : ℝ)
    (hs : 0 < s) {m : α → ℂ} :
    SouzaPointwiseMultiplier G s (1 : ℝ≥0∞) (1 : ℝ≥0∞)
      hs le_rfl ENNReal.one_ne_top m ↔
    SouzaPointwiseSelfsClass G s (1 : ℝ≥0∞) (1 : ℝ≥0∞)
      hs le_rfl ENNReal.one_ne_top m := ...
```

18.4. $selfs$ multipliers are bounded.

File: `GoodGrid/Multipliers/MultipliersareBounded.lean`
Imports (within the project): `GoodGrid/BesovAtoms.lean`, `GoodGrid/Multipliers/Definition.lean`, `GoodGrid/Multipliers/Besovspq.lean`

A function in a selfs class is automatically bounded. The proof is a pretty martingale argument (paper Proposition 18.3): testing g against the canonical atom of a cell Q and applying the L^p embedding of Proposition 5.1 to the product bounds the L^p average of g over Q, uniformly over all cells; since the cells generate the σ-algebra (G_8), Lévy's upward theorem lets these averages converge to $|g(x)|$ at almost every point, giving the sup bound. The formalization sharpens the paper in a direction the non-Archimedean theorems require: the constant is *independent of the tail level* t, so that multipliers known only on deep cells are still uniformly bounded.

Proposition 18.3 ($selfs$ classes are bounded)**.** *There is* $C_{11} = C_{11}(\lambda_1, \lambda_2, C_1, s, p, q)$ *such that for every* $t \in \mathbb{N}$ *and every* g *with* $|g|_{\mathcal{B}^{s,t}_{p,q,selfs}} <$

∞,

$$|g(x)| \le C_{11}\, |g|_{\mathcal{B}^{s,t}_{p,q,selfs}} \quad \textit{for } m\textit{-a.e. } x.$$

In particular $\mathcal{B}^s_{p,q,selfs} \subset L^\infty$ and this inclusion is continuous.

Formalization. In `GoodGrid/Multipliers/MultipliersareBounded.lean`. The pointwise estimate from a single tail bound is `souzaPointwiseSelfsTailBound_norm_ae_le`; the estimate in terms of the tail seminorm, with the explicit constant displayed in the code below, is `souzaPointwiseSelfsTailNorm_norm_ae_le`; and the qualitative L^∞ membership of the whole class is `souzaPointwiseSelfsTailClass_norm_ae_bounded`. This sharpens paper Proposition 18.3 in two respects: the estimate is proved for the tail seminorms $|\cdot|_{\mathcal{B}^{s,t}_{p,q,selfs}}$ with a constant *independent of t* (this uniformity is what the non-Archimedean arguments below require), and the formalization also provides a *canonical* variant (`SouzaPointwiseCanonicalSelfsTailBound`), in which g is only tested against products with canonical Souza's atoms, together with the bridges `Bound.toCanonical` and `SouzaPositivePointwiseSelfsTailBound.toCanonical`; the canonical variant is what the infinite positive-cone theorem below consumes.

```
theorem souzaPointwiseSelfsTailNorm_norm_ae_le
    (G : GoodGridSpace (α := α))
    (s : ℝ) (p q : ℝ≥0∞)
    (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)]
    (hs_lt_inv : s < (p.toReal)⁻¹)
    {t : ℕ} {m : α → ℂ}
    (hm : SouzaPointwiseSelfsTailClass G s p q hs hp hp_top t m) :
    ∀ᵐ x ∂G.grid.μ,
      ‖m x‖ ≤
        souzaBesovLpLocalEmbeddingConstant G s p q *
          (2 * souzaAmbientRestrictionMultiplierConstant G s p + 1) *
            souzaPointwiseSelfsTailNorm G s p q hs hp hp_top t m := ...
```

18.5. Non-Archimedean behaviour in $\mathcal{B}^{\beta}_{p,\tilde{q},selfs}$.

File: `GoodGrid/Multipliers/NonArchimedeanProperty.lean (statements exposed by NonArchimedeanPropertyPositiveStandalone.lean)`
Imports (within the project): `GoodGrid/Multipliers/Definition.lean, GoodGrid/BesovAtoms.lean, GoodGrid/Multipliers/Besovspq.lean, GoodGrid/Multipliers/MultipliersareBounded.lean, GoodGrid/PositiveCone.lean, WeakGrid/Scales.lean`

These are the deepest theorems of the repository (paper Proposition 18.4 and Remark 18.5). The phenomenon: if the supports of the multipliers g_i are *separated by the grid* — every cell carrying a nonzero coefficient of f meets essentially one support, which is what hypothesis A quantifies — then each atom of the representation of f "sees" only one g_i, and the cost of $(\sum_i g_i)f$

is governed by the *largest* relevant multiplier norm instead of their sum: the estimate behaves like an ultrametric bound, whence *non-Archimedean*. The level hypothesis B makes the separation scale-respecting: an atom may only meet a multiplier whose tail level has already been reached. The proof multiplies the representation of f atomwise, expands each product $g_i a_Q$ by the restriction lemma on the induced grid of Q, transmutes the resulting β-smooth representations down to smoothness s (transmutation again — here `WeakGrid/Scales.lean` provides the $\beta \to s$ rescaling), and reassembles; the bounded overlap of the supports is exactly what keeps the reassembled cost additive in the right way. The positive versions add the support and positivity conclusions of Remark 18.5, whose proofs rest on the support rigidity of Section 17; the infinite-index versions require, in addition, the pointwise theory: absolute convergence of $\sum_i g_i(x)$ on $\{f \neq 0\}$, with the limit identified through the compactness of representations of Section 6.

If $g_i \in M(\mathcal{B}^s_{p,q})$, the naive estimate is

$$\Big|\big(\sum_i g_i\big) f\Big|_{\mathcal{B}^s_{p,q}} \leq \Big(\sum_i |g_i|_{M(\mathcal{B}^s_{p,q})}\Big) |f|_{\mathcal{B}^s_{p,q}}.$$

Remarkably, when the supports of the g_i are separated by the grid, one gets a far better estimate.

Theorem 18.4 (Non-Archimedean property, finite version)**.** *Let $\beta > s$ and $\tilde{q} \in [1, \infty]$. There is $C_{12} = C_{12}(\lambda_1, \lambda_2, C_1, s, \beta, p, q, \tilde{q})$ with the following property. Let Λ be finite, g_i with $|g_i|_{\mathcal{B}^{\beta,t_i}_{p,\tilde{q},selfs}} < \infty$ for $i \in \Lambda$ and $t_i \in \mathbb{N}$. Consider a function f with a $\mathcal{B}^s_{p,q}$-representation*

$$f = \sum_k \sum_{Q \in \mathcal{P}^k} c_Q a_Q$$

by canonical Souza's atoms satisfying

A. *We have*

$$\sup_{\substack{Q \in \mathcal{P} \\ c_Q \neq 0}} \sum_{i:\ Q \cap \operatorname{supp} g_i \neq \emptyset} |g_i|_{\mathcal{B}^{\beta,t_i}_{p,\tilde{q},selfs}} \leq N.$$

B. *If $Q \in \mathcal{P}^k$ satisfies $c_Q \neq 0$ and $Q \cap \operatorname{supp} g_i \neq \emptyset$ then $k \geq t_i$.*

Then $(\sum_{i \in \Lambda} g_i) f \in \mathcal{B}^s_{p,q}$ and we can find a $\mathcal{B}^s_{p,q}$-representation

$$\big(\sum_{i \in \Lambda} g_i\big) f = \sum_k \sum_{Q \in \mathcal{P}^k} d_Q a_Q \tag{7}$$

by canonical Souza's atoms such that

$$\Big(\sum_k \big(\sum_{Q \in \mathcal{P}^k} |d_Q|^p\big)^{q/p}\Big)^{1/q} \leq C_{12}\, N \Big(\sum_k \big(\sum_{Q \in \mathcal{P}^k} |c_Q|^p\big)^{q/p}\Big)^{1/q}. \tag{8}$$

Formalization. The theorem `souzaNonArchimedeanPropertyLambdaFinite` in `GoodGrid/Multipliers/NonArchimedeanProperty.lean` (paper Proposition 18.4 for finite Λ) produces, from the bounded-overlap hypothesis A and the level/scale hypothesis B, a representation of the product $(\sum_i g_i)f$ of cost at most $C_{12}\,N$ times the cost of the representation of f, with C_{12} independent of the multipliers, of N, and of Λ. The representations of f and of the product are by *canonical* Souza's atoms (`SouzaCanonicalRepresentation`), which is no loss of generality for Souza's atoms and is the precise form formalized.

```
theorem souzaNonArchimedeanPropertyLambdaFinite
    (G : GoodGridSpace (α := α))
    (s β : ℝ) (p q qtilde : ℝ≥0∞)
    (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
    (hβ_lt_inv : β < (p.toReal)⁻¹)
    (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
    ∃ Cgen : ℝ,
      0 ≤ Cgen ∧
      ∀ (Λ : Finset ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
        (f : α → ℂ)
        (x : WeakGridSpace.BesovishSpace
          (souzaAtomFamily G s p hs hp hp_top) q)
        (R : WeakGridSpace.LpGridRepresentation
          (souzaAtomFamily G s p hs hp hp_top)
          (x : Lp ℂ p G.toWeakGridSpace.measure)),
          0 ≤ N →
          WeakGridSpace.RepresentsFunction
            (G := G.toWeakGridSpace) (p := p) f
            (x : Lp ℂ p G.toWeakGridSpace.measure) →
          WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
          (∀ i ∈ Λ,
            ∃ C : ℝ,
              SouzaPointwiseSelfsTailBound
                G β p qtilde hβ hp hp_top (t i) (g i) C) →
          (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
            (R.block k).coeff Q ≠ 0 →
              nonArchimedeanRelevantTailSelfsSum
                G β p qtilde hβ hp hp_top Λ t g Q ≤ N) →
          (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
            i ∈ Λ →
              (R.block k).coeff Q ≠ 0 →
                goodGridLevelCellMeetsSupport G Q (g i) →
                  t i ≤ k) →
          ∃ y : WeakGridSpace.BesovishSpace
              (souzaAtomFamily G s p hs hp hp_top) q,
            ∃ S : WeakGridSpace.LpGridRepresentation
                (souzaAtomFamily G s p hs hp hp_top)
                (y : Lp ℂ p G.toWeakGridSpace.measure),
              WeakGridSpace.RepresentsFunction
                (G := G.toWeakGridSpace) (p := p)
                (fun z => (∑ i ∈ Λ, g i z) * f z)
                (y : Lp ℂ p G.toWeakGridSpace.measure) ∧
              WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
```

```
              Cgen * N *
                WeakGridSpace.LpGridRepresentation.pqCost (q := q) R := ...
```

Theorem 18.5 (Non-Archimedean property, infinite version)**.** *In the setting of Theorem 18.4 let $\Lambda \subset \mathbb{N}$ be infinite, and replace hypothesis A. by the summability condition*

$$\sum_{i\in\Lambda} T_i \leq N, \qquad T_i := \sup_{\substack{Q\in\mathcal{P},\ c_Q\neq 0\\ Q\cap \operatorname{supp} g_i\neq\emptyset}} |g_i|_{\mathcal{B}^{\beta,t_i}_{p,\tilde{q},selfs}},$$

understood as a series with values in $[0,\infty]$. Then, for m-a.e. x in $\{f \neq 0\}$, the series $\sum_{i\in\Lambda} g_i(x)$ converges absolutely, with

$$\sum_{i\in\Lambda} |g_i(x)| \leq C_{12}\, N;$$

the function $h = (\sum_{i\in\Lambda} g_i)f$ (defined a.e. as this pointwise limit times f) belongs to L^p and to $\mathcal{B}^s_{p,q}$, and it has a canonical representation (7) satisfying (8).

Formalization. The theorem `souzaNonArchimedeanProperty` in `GoodGrid/Multipliers/NonArchimedeanProperty.lean` proves all the conclusions above: pointwise absolute summability of $\sum_i g_i(x)$ with bound $C_{12}N$ a.e. on $\{f \neq 0\}$, existence of the limit function h with $\|h\| \leq C_{12}N\|f\|$ a.e., membership of h in L^p, and a canonical representation of h with the cost bound (8). The pointwise part is `exists_nonArchimedeanInfinite_pointwise_hasSum`; the identification of the limit representation goes through the compactness of representations with uniform cost (`exists_strongly_convergent_subseq_of_uniform_pqCost`) applied to the truncations to finite initial segments of Λ. Paper Proposition 18.4 treats finite and infinite Λ together, reducing to the finite case; the formalization keeps them separate because the infinite case has genuinely more content.

```
theorem souzaNonArchimedeanProperty
    (G : GoodGridSpace (α := α))
    (s β : ℝ) (p q qtilde : ℝ≥0∞)
    (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
    (hβ_lt_inv : β < (p.toReal)⁻¹)
    (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
    ∃ Cgen : ℝ,
      0 ≤ Cgen ∧
      1 ≤ Cgen ∧
      ∀ (Λ : Set ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
        (f : α → ℂ)
        (x : WeakGridSpace.BesovishSpace
          (souzaAtomFamily G s p hs hp hp_top) q)
        (R : WeakGridSpace.LpGridRepresentation
          (souzaAtomFamily G s p hs hp hp_top)
          (x : Lp ℂ p G.toWeakGridSpace.measure)),
          0 ≤ N →
```

```
    WeakGridSpace.RepresentsFunction
      (G := G.toWeakGridSpace) (p := p) f
      (x : Lp ℂ p G.toWeakGridSpace.measure) →
    WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
    (∀ i ∈ Λ,
      ∃ C : ℝ,
        SouzaPointwiseSelfsTailBound
          G β p qtilde hβ hp hp_top (t i) (g i) C) →
    (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
      (R.block k).coeff Q ≠ 0 →
        ∃ T : ℝ,
          HasSum
            (fun i : {i // i ∈ Λ} =>
              nonArchimedeanRelevantTailSelfsInfiniteTerm
                G β p qtilde hβ hp hp_top Λ t g Q i)
            T ∧
          T ≤ N) →
    (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
      i ∈ Λ →
        (R.block k).coeff Q ≠ 0 →
          goodGridLevelCellMeetsSupport G Q (g i) →
            t i ≤ k) →
    ∃ h : α → ℂ,
      ∃ absSum : α → ℝ,
        (∀ᵐ z ∂G.toWeakGridSpace.measure,
          f z ≠ 0 →
            HasSum
              (fun i : {i // i ∈ Λ} => ‖g i.1 z‖)
              (absSum z) ∧
            absSum z ≤ Cgen * N) ∧
        (∀ᵐ z ∂G.toWeakGridSpace.measure,
          HasSum
            (fun i : {i // i ∈ Λ} => g i.1 z * f z)
            (h z)) ∧
        (∀ᵐ z ∂G.toWeakGridSpace.measure,
          ‖h z‖ ≤ Cgen * N * ‖f z‖) ∧
        (∃ hmem : MemLp h p G.toWeakGridSpace.measure,
          ‖MemLp.toLp h hmem‖ ≤
            Cgen * N * ‖(x : Lp ℂ p G.toWeakGridSpace.measure)‖) ∧
        ∃ y : WeakGridSpace.BesovishSpace
            (souzaAtomFamily G s p hs hp hp_top) q,
          ∃ S : WeakGridSpace.LpGridRepresentation
              (souzaAtomFamily G s p hs hp hp_top)
              (y : Lp ℂ p G.toWeakGridSpace.measure),
            WeakGridSpace.RepresentsFunction
              (G := G.toWeakGridSpace) (p := p) h
              (y : Lp ℂ p G.toWeakGridSpace.measure) ∧
            WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) S ∧
            WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
              Cgen * N *
                WeakGridSpace.LpGridRepresentation.pqCost (q := q) R := ...
```

Theorem 18.6 (Non-Archimedean property, positive version)**.** *In the setting of Theorems 18.4 and 18.5 (finite or infinite Λ, with the respective hypotheses, the tail seminorms being replaced by their positive-cone counterparts*

$$|g|_{\mathcal{B}^{\beta+,t}_{p,\tilde{q},selfs}} = \sup_{\substack{a_Q \in \mathcal{A}^{sz}_{\beta,p} \\ Q \in \mathcal{P}^j,\ j \geq t}} |g a_Q|_{\mathcal{B}^{\beta+}_{p,\tilde{q}}}$$

for the functions g_i), assume that each g_i is $\mathcal{B}^{\beta,t_i}_{p,\tilde{q}}$-positive. Let R be the given canonical representation $(c_Q)_Q$ of f. Then the representation $S = (d_Q)_Q$ of $(\sum_i g_i)f$ produced by Theorems 18.4/18.5 can be chosen so that, in addition:

i. *(positivity) if R is a positive representation ($c_Q \geq 0$ for all Q), then S is a cone-positive representation: all d_Q are nonnegative reals and the corresponding Souza's atoms are a.e. nonnegative;*
ii. *(support) unconditionally — without assuming $c_Q \geq 0$ — every cell Q active in S ($d_Q \neq 0$) satisfies: there exists $i \in \Lambda$ with $g_i \neq 0$ a.e. on Q.*

Formalization. In the file `NonArchimedeanPropertyPositiveStandalone.lean` of `GoodGrid/Multipliers/`, with cores in `NonArchimedeanProperty.lean` and `GoodGrid/PositiveCone.lean`. The finite case is the theorem `souzaNonArchimedeanPropertyPositiveCone`: it produces the representation S of the product with the cost bound of Theorem 18.4 and, in addition, the support conclusion ii. and the positivity implication i. The infinite case is `souzaNonArchimedeanPropertyPositiveConeInfinite` ($\Lambda \subset \mathbb{N}$ infinite, with condition A as an $[0,\infty]$-valued series bound, no summability witness needed): it proves all the conclusions of Theorem 18.5 plus i. and ii.; the limit arguments pass ii. through the convergence of the coefficients and i. through the closedness of the nonnegative real ray in $\mathbb{C}$ and a.e. convergence of a subsequence of the atoms. The auxiliary product representation behind both is `exists_nonArchimedeanProductRepresentation_positive`, whose key local ingredient (`exists_nonArchimedeanLocalTransmutationData_pos`) assembles, one multiplier at a time, positive local transmutation data on the induced grids. Compared with paper Remark 18.5 the statement was made precise as follows: (a) the hypothesis on the representation R of f is only that it is canonical (arbitrary complex coefficients); (b) the two conclusions are separated according to their true strengths — the support conclusion ii. holds unconditionally, and is weakened to its a.e. form ($g_i \neq 0$ a.e. on Q rather than $Q \subset \operatorname{supp} g_i$); (c) the positivity conclusion i. produces a *cone-positive* representation (nonnegative real coefficients, a.e. nonnegative Souza's atoms), not necessarily one by canonical atoms. Axioms of all four non-Archimedean theorems checked: `propext`, `Classical.choice`, `Quot.sound`.

```
theorem souzaNonArchimedeanPropertyPositiveCone
    (G : GoodGridSpace (α := α))
    (s β : ℝ) (p q qtilde : ℝ≥0∞)
    (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
    (hβ_lt_inv : β < (p.toReal)⁻¹)
```

```
  (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
∃ Cgen : ℝ,
  0 ≤ Cgen ∧
  ∀ (Λ : Finset ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
    (f : α → ℂ)
    (x : WeakGridSpace.BesovishSpace
      (souzaAtomFamily G s p hs hp hp_top) q)
    (R : WeakGridSpace.LpGridRepresentation
      (souzaAtomFamily G s p hs hp hp_top)
      (x : Lp ℂ p G.toWeakGridSpace.measure)),
      0 ≤ N →
      WeakGridSpace.RepresentsFunction
        (G := G.toWeakGridSpace) (p := p) f
        (x : Lp ℂ p G.toWeakGridSpace.measure) →
      WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
      SouzaCanonicalRepresentation G s p hs hp hp_top R →
      (∀ i ∈ Λ,
        SouzaPositiveFunction G β p qtilde hβ hp hp_top (g i)) →
      (∀ i ∈ Λ,
        ∃ C : ℝ≥0∞,
          SouzaPositivePointwiseSelfsTailBound
            G β p qtilde hβ hp hp_top (t i) (g i) C) →
      (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
        (R.block k).coeff Q ≠ 0 →
          nonArchimedeanRelevantPositiveTailSelfsSum
            G β p qtilde hβ hp hp_top Λ t g Q ≤ ENNReal.ofReal N) →
      (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
        i ∈ Λ →
          (R.block k).coeff Q ≠ 0 →
            goodGridLevelCellMeetsSupport G Q (g i) →
              t i ≤ k) →
      ∃ y : WeakGridSpace.BesovishSpace
          (souzaAtomFamily G s p hs hp hp_top) q,
        ∃ S : WeakGridSpace.LpGridRepresentation
            (souzaAtomFamily G s p hs hp hp_top)
            (y : Lp ℂ p G.toWeakGridSpace.measure),
          WeakGridSpace.RepresentsFunction
            (G := G.toWeakGridSpace) (p := p)
            (fun z => (∑ i ∈ Λ, g i z) * f z)
            (y : Lp ℂ p G.toWeakGridSpace.measure) ∧
          WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) S ∧
          WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
            Cgen * N *
              WeakGridSpace.LpGridRepresentation.pqCost (q := q) R ∧
          -- [ii] support: holds with only canonical atoms on `R` (no sign on
 `c_Q`).
          (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
            (S.block k).coeff Q ≠ 0 →
              ∃ i ∈ Λ, ∀ᵐ z ∂(G.toWeakGridSpace.measure.restrict Q.1), g i z
 ≠ 0) ∧
          -- [i] positivity: only when `R` is fully positive (canonical + `c_Q
 ≥ 0`).
          (SouzaPositiveRepresentation G s p hs hp hp_top R →
            SouzaConePositiveRepresentation G s p hs hp hp_top S) := ...
```

```
theorem souzaNonArchimedeanPropertyPositiveConeInfinite
    (G : GoodGridSpace (α := α))
    (s β : ℝ) (p q qtilde : ℝ≥0∞)
    (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
    (hβ_lt_inv : β < (p.toReal)⁻¹)
    (hp : 1 ≤ p) (hp_top : p ≠ ∞)
    [Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
    ∃ Cgen : ℝ,
      0 ≤ Cgen ∧
      1 ≤ Cgen ∧
      ∀ (Λ : Set ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
        (f : α → ℂ)
        (x : WeakGridSpace.BesovishSpace
          (souzaAtomFamily G s p hs hp hp_top) q)
        (R : WeakGridSpace.LpGridRepresentation
          (souzaAtomFamily G s p hs hp hp_top)
          (x : Lp ℂ p G.toWeakGridSpace.measure)),
          0 ≤ N →
          WeakGridSpace.RepresentsFunction
            (G := G.toWeakGridSpace) (p := p) f
            (x : Lp ℂ p G.toWeakGridSpace.measure) →
          WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
          SouzaCanonicalRepresentation G s p hs hp hp_top R →
          (∀ i ∈ Λ,
            SouzaPositiveFunction G β p qtilde hβ hp hp_top (g i)) →
          (∀ i ∈ Λ,
            ∃ C : ℝ≥0∞,
              SouzaPositivePointwiseSelfsTailBound
                G β p qtilde hβ hp hp_top (t i) (g i) C) →
          (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
            (R.block k).coeff Q ≠ 0 →
              (∑' i : {i // i ∈ Λ},
                nonArchimedeanRelevantPositiveTailSelfsInfiniteTerm
                  G β p qtilde hβ hp hp_top Λ t g Q i) ≤ ENNReal.ofReal N) →
          (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
            i ∈ Λ →
              (R.block k).coeff Q ≠ 0 →
                goodGridLevelCellMeetsSupport G Q (g i) →
                  t i ≤ k) →
          ∃ h : α → ℂ,
            ∃ absSum : α → ℝ,
              (∀ᵐ z ∂G.toWeakGridSpace.measure,
                f z ≠ 0 →
                  HasSum
                    (fun i : {i // i ∈ Λ} => ‖g i.1 z‖)
                    (absSum z) ∧
                  absSum z ≤ Cgen * N) ∧
              (∀ᵐ z ∂G.toWeakGridSpace.measure,
                HasSum
                  (fun i : {i // i ∈ Λ} => g i.1 z * f z)
                  (h z)) ∧
              (∀ᵐ z ∂G.toWeakGridSpace.measure,
                ‖h z‖ ≤ Cgen * N * ‖f z‖) ∧
              (∃ hmem : MemLp h p G.toWeakGridSpace.measure,
```

```
          ‖MemLp.toLp h hmem‖ ≤
            Cgen * N * ‖(x : Lp ℂ p G.toWeakGridSpace.measure)‖) ∧
        ∃ y : WeakGridSpace.BesovishSpace
            (souzaAtomFamily G s p hs hp hp_top) q,
          ∃ S : WeakGridSpace.LpGridRepresentation
              (souzaAtomFamily G s p hs hp hp_top)
              (y : Lp ℂ p G.toWeakGridSpace.measure),
            WeakGridSpace.RepresentsFunction
              (G := G.toWeakGridSpace) (p := p) h
              (y : Lp ℂ p G.toWeakGridSpace.measure) ∧
            WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) S ∧
            WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
              Cgen * N *
                WeakGridSpace.LpGridRepresentation.pqCost (q := q) R ∧
            (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
              (S.block k).coeff Q ≠ 0 →
                ∃ i ∈ Λ,
                  ∀ᵐ z ∂(G.toWeakGridSpace.measure.restrict Q.1),
                    g i z ≠ 0) ∧
            (SouzaPositiveRepresentation G s p hs hp hp_top R →
              SouzaConePositiveRepresentation G s p hs hp hp_top S) := ...
```

Remark 18.7. Paper Corollary 18.6 ($\mathcal{B}^{\beta}_{p,\tilde{q},selfs} \subset M(\mathcal{B}^{s}_{p,q})$ continuously, for $\beta > s$) follows from the formalized Theorem 18.4 exactly as in the paper, but it is not itself stated in the repository; it is therefore omitted from this document.

III. GUIDE TO THE REPOSITORY

19. Guide to the repository files

All paths below are relative to the library root `BesovSpacesGoodGrid/`; the aggregate file `BesovSpacesGoodGrid.lean` is the main entry point and imports the whole pipeline.

19.1. Weak-grid layer (Part I).

- `WeakGrid/Definition.lean`: weak-grid structures (`WeakGrid`, `WeakGridSpace`) and the overlap estimates of Section 3.
- `WeakGrid/Atoms.lean`: weak-grid cells, local spaces, and atom families (Section 4).
- `WeakGrid/BesovishSpaces.lean`: level blocks, representations, costs, Besov-ish spaces, the cost gauge `Norm_Costpq`, the L^t embedding and the normed structure (Section 5).
- `WeakGrid/Scales.lean`: scaled atom families and the smoothness-scale inclusion (Section 7).

- `WeakGrid/Completeness.lean`: representation limits, compactness of cost balls, and completeness (Section 6).
- `WeakGrid/InducedGrid.lean`: induced weak grids on a fixed cell and the contraction into the ambient Besov-ish space (Section 8).
- `WeakGrid/Transmutation.lean`: transmutation objects and Claims I, II, III, including the explicit Claim C embedding constants (Section 9).
- `WeakGrid/Multipliers.lean`: the abstract multiplier and selfs-class predicates on weak grids (Section 10).
- `Sums.lean`: block-index types (`BlockIndex`, `ChildIndex`, `ParentIndex`) and block-sum notation used throughout (Section 10).

19.2. **Good-grid layer (Part II).**

- `GoodGrid/Definition.lean`: good-grid structures (Section 11).
- `GoodGrid/BesovSpace.lean`: Souza's atoms and good-grid Besov spaces, with completeness, compactness, and density (Sections 12 and 13).
- `GoodGrid/PositiveCone.lean`: the positive cone, positive and cone-positive representations, decompositions, density, and the support theorems (Section 17).
- `GoodGrid/BesovAtoms.lean`: Besov's atoms on good grids and the Souza/Besov sandwich theorem (Section 14).
- `GoodGrid/AlternativeRepresentationsAndNorms.lean` and `GoodGrid/AlternativeRepresentationsAndNorms/`: the norm-comparison layer of Section 16 — `HaarRepresentationNorm.lean` (normalized Haar coefficients and the Haar gauge), `HaarParametrizedRepresentation.lean` and `ComparingHaarRepresentationsl.lean` (Haar expansions as parametrized representations and their comparison), `standardRepresentation.lean` (standard atomic coefficients and gauge), `standardNormleqHaarRepresenstionNorm.lean` (Haar controls standard), `StandarRepresentationNormleqBesovNorm.lean` (the Besov norm controls the standard norm), `FiniteStandardNormimpliesBesov.lean` (finite standard norm endpoint), `FiniteHaarNormimpliesLp.lean` (finite Haar norm endpoint), `MeanOscillationNorm.lean` (mean-oscillation definitions and local oscillation lemmas), `OscillationNormleqBesovNorm.lean` (Besov controls mean oscillation), and `HaarNormleqOscillationNorm.lean` (mean oscillation controls Haar).

- `GoodGrid/Distribution.lean`: cell indicators (`atomIndicator`), the test-function module (`TestFunctions`), and its algebraic dual (`Distributions`).

19.3. Multiplier layer (Part II, Section 18).

- `GoodGrid/Multipliers.lean` and `GoodGrid/Multipliers/`: the pointwise-multiplier layer of Section 18 — `Definition.lean` (the `selfs` classes and seminorms), `Besovspq.lean` (induced-cell representations, restriction to a cell, and the general inclusion $M(\mathcal{B}^s_{p,q}) \subset \mathcal{B}^s_{p,q,selfs}$), `Besovs11.lean` (the $\mathcal{B}^s_{1,1}$ characterization), `MultipliersareBounded.lean` (boundedness of `selfs` multipliers, uniform in the tail level, and the canonical tail classes), `NonArchimedeanProperty.lean` (the finite and infinite non-Archimedean estimates and the cores of their positive versions), and `NonArchimedeanPropertyPositiveStandalone.lean` (the user-facing positive-cone statements, finite and infinite).

19.4. **External dependencies.** The repository depends on the external Lean repositories `UnbalancedHaarWavelet` (grids and unbalanced Haar systems), `UnconditionalSchauderBasis`, `LaminarFamiliesMaximalBinaryTrees` and `Burkholder` (all by SmaniaD), which provide the Girardi–Sweldens construction and the unconditional basis property used in Section 15.

19.5. **Repository status.** At the date of this document (June 9, 2026) a full `lake build` succeeds (3454 jobs) and the project Lean files contain no `sorry`, no `admit`, and no project-local `axiom` or `constant` declarations. The axioms of the main theorems — the finite and infinite, non-positive and positive non-Archimedean properties, namely

`souzaNonArchimedeanProperty`,
`souzaNonArchimedeanPropertyLambdaFinite`,
`souzaNonArchimedeanPropertyPositiveCone`,
`souzaNonArchimedeanPropertyPositiveConeInfinite`

— were checked with `#print axioms`: only the standard axioms `propext`, `Classical.choice`, `Quot.sound` appear (no `sorryAx`). The running verification log is kept in `status.md` at the repository root.

References

[1] D. Smania, *Besov-ish spaces through atomic decomposition*, Analysis & PDE **15** (2022), no. 1, 123–174. A prerelease version of the published paper is included in this repository as `preprint-source/besovish-2020-07-18.tex` (with PDF); it is the source from which the present document was produced.

[2] M. Girardi and W. Sweldens, *A new class of unbalanced Haar wavelets that form an unconditional basis for L_p on general measure spaces*, J. Fourier Anal. Appl. **3** (1997), 457–474.

[3] G. S. de Souza, *Spaces formed by special atoms*, PhD thesis, SUNY Albany, 1980.

[4] D. Smania, *BesovSpacesGoodGrid: a Lean 4 formalization*, git repository, 2026. Dependencies: `UnbalancedHaarWavelet`, `UnconditionalSchauderBasis`, `LaminarFamiliesMaximalBinaryTrees`, `Burkholder` (SmaniaD).

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